Costs of Adjustment and Firms' Responsiveness: Evidence from a Labour Market Reform

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Abstract

Following the European debt crisis various countries added more flexibility to their labour markets in an attempt to boost productivity. However, in Portugal the distribution of labour productivity has remained unchanged even after a comprehensive reform effectively slashed labour adjustment costs. Why did this happen? Is productivity immune to changes in adjustment costs which constrain firms' labour demand decisions? I study this puzzling development with a dataset of Portuguese firms. While relying on a revenue function estimation exercise I identify the unobserved profitability shocks in the data and show that firms actually respond more to those exogenous innovations in the period following the reform: the responsiveness coefficient in the employment policy function nearly doubled. Importantly, I also document a significant fall in the curvature of firms' revenue function in the post-reform period. As the marginal worker does not produce as much revenue, increasing employment responsiveness fails to be translated in productivity gains. To corroborate this point I develop and estimate a structural model of firm dynamics. A counterfactual exercise demonstrates that, although in opposite directions, reduced costs of adjustment and a falling curvature exert quantitatively similar effects on the variance of labour productivity, a measure linked to misallocation. The flat dispersion observed in the data emerges as a tension between these channels: a more flexible labour market enhances firms' employment responsiveness to shocks but the corresponding effects on labour productivity are offset by a reduction in the curvature of their revenue function.

JEL codes: E20, E24, J20, J23, L22, L25

Keywords: Productivity, Responsiveness, Labour adjustment costs, Misallocation

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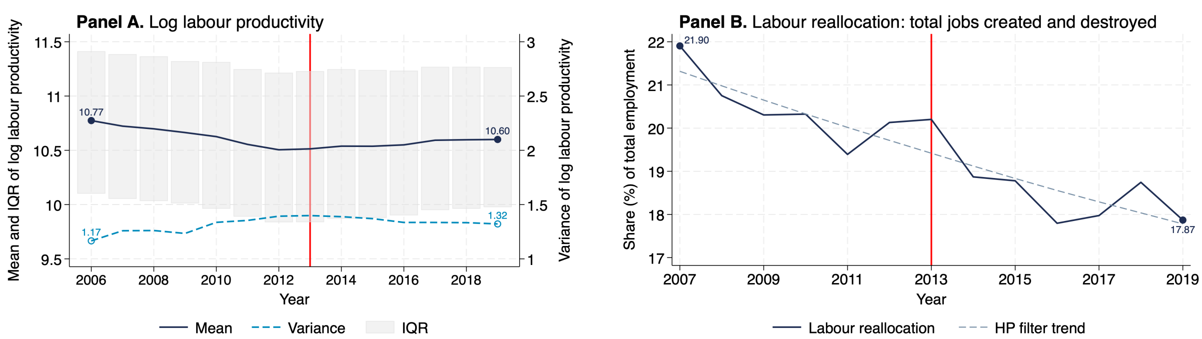
1 Introduction

Slowing growth and flat productivity have been ailing most European economies since the start of the century. Naturally, pressure has mounted on policy-makers to design sound and effective strategies which can reverse these trends. This paper studies the productivity effects of a comprehensive reform which reduced labour adjustment costs borne by Portuguese firms. The context is particularly interesting. In 2008, prior to the enacted changes, the labour market in Portugal featured as the most stringent across all OECD countries. Owing to significant reductions in the amount of severance paid to dismissed employees and the introduction of a new and more flexible reason to justify dismissals, amongst numerous other provisions, costs of firing were slashed. Ideally, these looser constraints would free firms to be more responsive to shocks hitting their profitability. This would then speed up the labour reallocation process from low- to more-profitable firms, and thus contribute to raise aggregate productivity in the economy.

This was arguably the ultimate goal the reform attempted to accomplish. Shortly before the first set of measures was introduced, for example, the Portuguese government itself acknowledged the need to promote *"an accelerated increase of productivity"* in an agreement signed with employers' associations and labour unions. A similar assessment would be found years later in a report commissioned to evaluate the impact of those policies: OECD (2017) posited very clearly that *"cuts in severance pay are expected to result in significant gains in both productivity and growth"*. Yet these gains have been nonexistent. Although new contracts have been under the umbrella of more flexible clauses since late 2013, neither the cross-sectional mean nor the dispersion of log labour productivity have exhibited noticeable swings (Figure 1, panel A). These also seem to have been incapable of reversing the declining pace of job flows across firms (Figure 1, panel B).

Figure 1: Recent trends in the distribution of (log) labour productivity and labour reallocation



Notes: *Labour productivity* is measured as the ratio of a firm's deflated revenues to their stock of workers. The cross-sectional mean and interquartile range (IQR) are displayed on the left axis, the variance on the right. *Labour reallocation* is the sum of jobs created by entering and expanding firms, and jobs destroyed by exiting and contracting firms. The fitted trend relies on an HP filter with smoothing parameter 100. Data from Banco de Portugal Microdata Research Laboratory (2025).

Despite the increasing flexibility added to the labour law, the productivity distribution remained largely unchanged. **Why did this happen? Is productivity surprisingly immune to changes in adjustment costs which constrain firms' labour demand choices?** A potential answer to these questions resides in the concept of employment responsiveness: if, contrary to what had been expected, these lower costs of adjustment did not prompt firms to respond more to changes in their profitability circumstances, then the efficient labour reallocation process would be kept impaired and be a drag on productivity. I discard this hypothesis with the aid of three facts taken from a rich dataset of Portuguese firms. First, an estimated log linear approximation of the employment policy function reveals that the responsiveness coefficient nearly doubles in the post-reform period. This is consistent with additional reduced-form evidence: both the covariance and the correlation coefficient between profitability and employment rise in the period following the reform. Second, I show that this enhanced responsiveness can be found at the extensive and intensive margins of labour adjustment. I report the existence of a convex adjustment hazard function and a concave hiring rule, similar to that of Ilut et al. (2018), which becomes more linear in the presence of lower costs of adjustment. And third, I demonstrate that rising responsiveness is not matched by an increasing probability of exit when poor profitability innovations are drawn.

As the reform loosened labour adjustment costs, employment responsiveness rose. This is a novel and particularly relevant result to an ongoing debate on the link between productivity, business dynamism, and firms' responsiveness. Decker et al. (2016) and Decker et al. (2020) have supported the hypothesis that the well-documented decline in the pace of job reallocation in the US is explained by a weaker responsiveness of firms to exogenous shocks. While Cooper et al. (2024a) have subsequently claimed this is essentially the reflex of increased costs of labour adjustment, De Loecker et al. (2020) have instead pointed out waning business dynamism is justified by changes in market structures and specifically a rise in firms' market power. My contribution is twofold. First, I prove that Portuguese firms have recently become more responsive to changes in their profitability. This conclusion stands in contrast with that of Decker et al. (2020), but also that by Biondi et al. (2026) for most European countries. And second, I demonstrate this finding is mostly the product of lower labour adjustment costs, a view consistent with that of Cooper et al. (2024a).

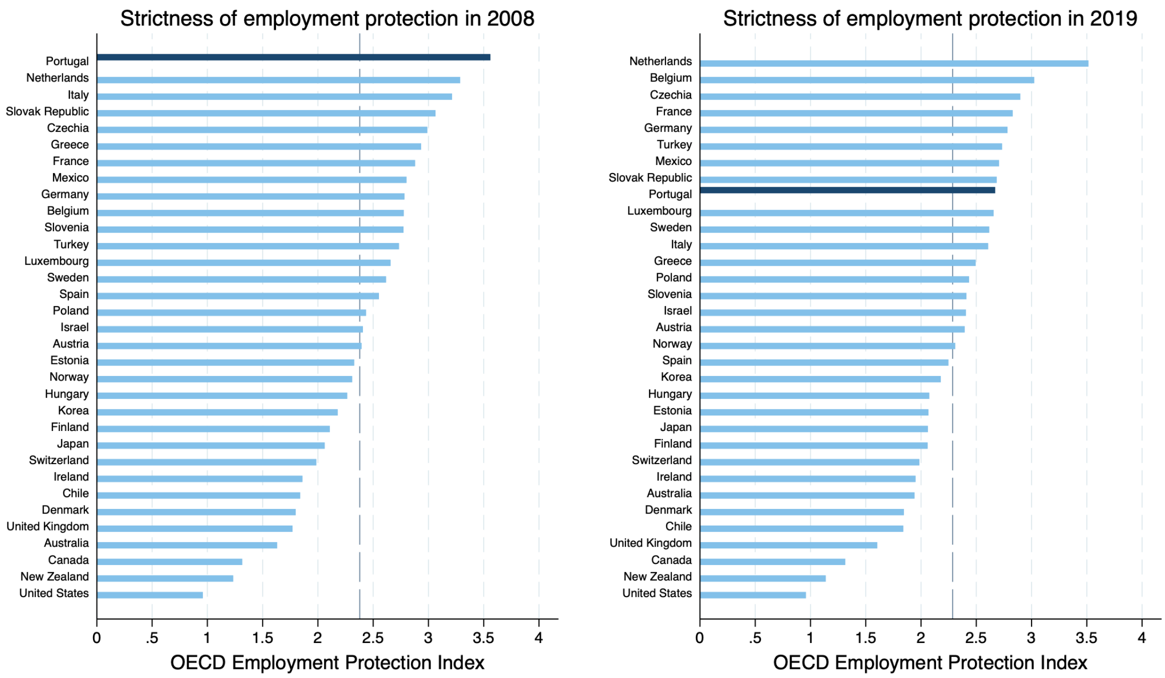
The ability to establish this enhanced responsiveness hinges on a revenue function estimation exercise which retrieves unobserved profitability innovations from the data. The usefulness of such a procedure actually expands beyond the identification of these exogenous shocks. Importantly, my GMM-IV estimates indicate that the period following the reform coincided with a statistically significant fall in the curvature of firms' revenue function. By reducing the revenue generated by the marginal worker this can sever the link from increasing responsiveness to productivity gains.

The existence of a nearly constant labour productivity variance merits particular attention here given its connection to the concept of misallocation, *i.e.* the dispersion of TFPR, defined in the seminal work of Hsieh and Klenow (2009). **Does that flat dispersion reflect the interaction between a more flexible labour market and changes in the curvature of the revenue function?** An answer to this question requires that the productivity effects of reduced costs of adjustment and those of a falling curvature be separately identified. I do so in a canonical model of firm dynamics in the tradition of Hopenhayn (1992) and Hopenhayn and Rogerson (1993), and which I estimate through simulated method of moments. In my partial equilibrium framework idiosyncratic shocks to profitability induce firms to alter their employment decisions and, in some cases, to endogenously exit the economy. Albeit simple, the model retains very useful properties. First, it allows to rationalise the magnitude of these responses by appealing to three components which can be disentangled: *i)* a piecewise formulation of convex and non-convex labour adjustment costs, *ii)* the demand structure embedded in the curvature of firms' revenue function, and *iii)* the stochastic process which governs their profitability. And second, it also allows to express moments of the labour productivity distribution as functions of these three elements.

I highlight two main results delivered by the model. First, the dispersion of labour productivity declines when adjustment costs are cut, but it increases when the curvature of the revenue function falls. In a counterfactual exercise I show that if the curvature had not fallen, the standard deviation of labour productivity would have dropped 14%. Conversely, if labour adjustment costs had not been slashed, that standard deviation would have actually risen 11%. Such similar magnitudes suggest the two mechanisms do exert a quantitatively neutral impact on the dispersion of labour productivity, a tension which is present in the data. And second, although the distribution is left unchanged, decomposing its variance does reveal rich dynamics. The parameters governing the revenue function play a pivotal role here: they fully determine the increase in the misallocation measure, and they also raise the weighted variance of employment in the cross-section.

Ultimately, the paper argues that lower costs of adjustment in the Portuguese economy did enhance firms' employment responsiveness but the corresponding effect on productivity was fully offset by a concurrent reduction in the curvature of their revenue function. Although I focus on Portugal, these conclusions can be farther extended. Larger countries like Italy and Spain also introduced more flexibility to their national labour laws in an attempt to boost productivity and revamp their economies. But as widespread as these policies might have been, a complete assessment of their productivity consequences is still missing. This is especially important now that muted growth and the lack of competitiveness have retaken centre stage in European policy circles. The paper contributes to that debate by drawing lessons from a reform which successfully added flexibility to the most rigid labour market in the OECD — see Figure 2. This is a non-negligible development, which can be largely attributed to the multiple changes lawmakers introduced to the Portuguese labour law between 2011 and 2013. A comprehensive overview of all those, as well as their intended goals, can be found in OECD (2017). Yet my emphasis is placed on three very specific policies which aimed to reduce the costs borne by firms when letting go employees: *i)* severance payments were slashed from 30 days of base wage for every full year of tenure at the firm to only 12 days, *ii)* firm-worker funds were set up in order to accommodate up to half of severance due in case of dismissal, and *iii)* a new and more flexible justification for fair dismissals was created. In Appendix A I go over their institutional details.

Figure 2: OECD Employment protection legislation index in 2008 (left) and 2019 (right)



Notes: Data from OECD Employment Protection Legislation Index, version 3. Greater/smaller values of the index are understood as more/less rigid labour markets. Only OECD countries with 2008 data available were considered. Gray dashed line is the cross-section unweighted average.

The paper also speaks to literature pioneered by Hamermesh (1989), Bentolila and Bertola (1990), and Caballero et al. (1997) which studies firms' labour demand decisions in the presence of adjustment costs, and how these impact the macroeconomy. Hopenhayn and Rogerson (1993) showed that a linear cost of job destruction can reduce average productivity by more than 2%. I rely on an adjustment cost function which supplements theirs. My piecewise formulation distinguishes net costs of job creation from job destruction, and includes both fixed and linear costs of adjustment. This specific coexistence of non-convex and convex adjustment costs is meant to capture prominent features of the Portuguese labour market (pervasive inaction, as documented by Varejão and Portugal (2007)) and targets of the reform (cut in severance payments, whose effectiveness has already been the subject of a preliminary analysis by Martins (2019)).

Lastly, I bridge this literature with another which assesses the effects of labour market reforms. Autor et al. (2007), Cingano et al. (2016), and Caggese et al. (2022) have all relied on purely empirical exercises to evaluate the impact of policy interventions in the US, Italy, and Belgium, respectively, which strengthen workers' rights at the expense of mobility and flexibility in the labour market. Cooper et al. (2018), on the other hand, also estimated a general equilibrium model of labour demand to explore the effects of a similar reform in China. Eventually, they all agree rising firing costs have a negative impact on aggregate employment and productivity. This is not entirely the case in Portugal. While studying a reform of the opposite sign in a highly stringent labour market, I show that the distribution of labour productivity actually remained unaltered even when real economic growth picked up.

Road map

The paper is organised as follows. In Section 2 I introduce the dataset and document a fall in the curvature of firms' revenue function. From this estimation procedure I also retrieve the unobserved profitability innovations which are used to explore the effects of the reform in different responsiveness regressions. The partial equilibrium model of labour demand with adjustment costs is presented in Section 3 and estimated in Section 4. In Section 5 I rely on counterfactual scenarios and a decomposition exercise to quantitatively assess the impact of the various channels on the variance of labour productivity. Section 6 concludes.

2 Data facts

Lower costs of labour adjustment prompt firms to respond more to changes in their profitability. By speeding up the reallocation of workers throughout the economy, this enhanced responsiveness can have a sizeable impact on labour productivity. In this section I leverage on a rich Portuguese dataset to study the evolution of employment responsiveness.

The annual data are provided by Banco de Portugal Microdata Research Laboratory (2025), an autonomous unit within the Bank of Portugal's Economics and Research Department, and comprise all balance sheet and income statement items of non-financial firms operating in Portugal. The dataset is based on information these firms are required to report by law. As a result, the coverage is extraordinary: the near-universe of non-financial firms is included. Table C1 displays some summary statistics.

Given the staggered implementation of the policies there is not a unique discontinuity which helps to causally disentangle their effects. For the remainder of the paper let the pre-reform period be 2006-2013, and the post-reform period 2014-2019. The data are available since 2006 and choosing 2019 as the final year avoids contamination from the covid shock. The choice of 2014 as a cut-off year is justified by two reasons. First, all policy measures described in the last section were implemented until late 2013. And second, because that coincides with the peak of employment inaction observed in the economy.[1] The 2014-2019 time window also retains a sufficiently large number of observations.

2.1 Revenue function estimation

Consider an economy where heterogeneous firms operate in monopolistic competition. The *real* side of the economy is described by a Cobb-Douglas production function: $y_{i,t} = z_{i,t} \cdot (n_{i,t}h_{i,t})^{\zeta} \cdot k_{i,t}^{1-\zeta}$, where $z_{i,t}$ is the idiosyncratic productivity level of firm i in period t, $n_{i,t}$ is the stock of employees, $h_{i,t}$ is the average number of hours worked, and $k_{i,t}$ is the stock of capital. For a reason I will discuss later, this formulation assumes constant returns to scale. On the *nominal* side, say these firms face a downward sloping demand curve given by $p(y_{i,t}) = (\theta_{i,t}/y_{i,t})^{1/\nu}$, where $\theta_{i,t}$ is a stochastic demand shifter, and ν the price elasticity of demand.

Under fully flexible capital, the revenue function simplifies to $R_{i,t} = A_{i,t} \cdot (n_{i,t}h_{i,t})^{\alpha}$. In Appendix B I show how to derive this closed form function from the primitives indicated above. Two elements are worth noticing. First, the idiosyncratic profitability level, $A_{i,t}$, encompasses shocks to productivity, demand, and the cost of capital. The literature commonly refers to it as a TFPR measure. It evolves according to a stochastic AR(1) process, such that $A_{i,t} = \xi_{i,t} \cdot A_{i,t-1}^{\rho}$ with $|\rho| < 1$ and $\log(\xi_{i,t}) \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$. And second, the curvature $\alpha \in (0,1)$ encompasses both the labour elasticity in the production function, $\zeta \in (0,1)$, and the price elasticity of demand, $\nu \geq 0$.

While capital and hours can be freely adjusted, the number of employees cannot. Regardless of the specification of labour adjustment costs, the employment policy function can be conceived as $n_{i,t} \equiv \varphi(n_{i,t-1}, \xi_{i,t})$. The past stock of workers is relevant to the decision because it is informative about the costs firms bear with potential adjustments. And since they also factor in their contemporaneous technological and demand conditions, the profitability innovation, $\xi_{i,t}$, is included as a second argument.

Despite their importance to firms' employment decisions, these innovations are unobserved to the researcher. But they can be retrieved from the data once the parameters which characterise the revenue function are estimated. To do so I apply logs and quasi first-difference it:

$$\log(R_{i,t}) = \rho \log(R_{i,t-1}) + \alpha\big(\log(n_{i,t}) + \log(h_{i,t})\big) - \alpha\rho\big(\log(n_{i,t-1}) + \log(h_{i,t-1})\big) + \log(\xi_{i,t}) \tag{1}$$

Estimating (1) via OLS would yield a biased estimator of α. Recall that from the policy function laid out before employment decisions are contemporaneously correlated with profitability innovations. The fact that $\log(n_{i,t}) \not\perp \log(\xi_{i,t})$ implies the existence of omitted variable bias. Instead, I rely on a set of instrumental variables to estimate (1) through GMM. The existence of adjustment costs adds some sluggishness to labour demand decisions. One-period lagged wages and effective labour usage must then be correlated with today's

[1] A useful timeline of events is provided in OECD (2017), p.155. For the evolution of employment inaction see Figure 3.

stock of workers. And by the definition of an i.i.d. innovation, the former must also be orthogonal to $\log(\xi_{i,t})$. Finally, a full set of year dummies controls for the effects of aggregate shocks. Table 1 shows the outcome of the estimation.

Table 1: GMM-IV revenue function estimation

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Curvature, α	0.526^{***}	0.563^{***}	0.442^{***}
	(0.005)	(0.015)	(0.017)
Serial correlation, ρ	0.880^{***}	0.878^{***}	0.901^{***}
	(0.003)	(0.004)	(0.008)
Std deviation, σ	0.601^{***}	0.601^{***}	0.594^{***}
	(0.005)	(0.015)	(0.017)
Observations	2 612 592	1 377 835	1 043 986

Notes: Parameters α and ρ are estimated directly from equation (1). The corresponding standard errors are clustered at the industry level. The series of profitability innovations are then generated with the parameterised revenue function. The estimator of σ is the standard deviation of that data series. The variance of this estimator is obtained by sampling with replacement 100 000 observations 100 times.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

While the serial correlation of TFPR and the standard deviation of profitability innovations hardly change throughout the years, the estimated curvature of the revenue function ranges between 0.44 (post-reform) and 0.56 (pre-reform). This is a statistically significant fall which does not rest on a changing composition of the economy — see Tables B1 and B2. It is also a result embedded with economic significance: since the marginal worker does not generate as much revenue, rising employment responsiveness fails to produce productivity gains. Although it is not the point of this paper to ascertain the causes of this reduction, in sub-section 2.5 I discuss and provide tentative evidence on the forces which may explain such a large change in parameter α.

In the sub-sections which follow immediately I rely on a useful by-product of the revenue function estimation exercise. Since the procedure uncovers the series of exogenous firm-specific profitability innovations, $\{\log(\xi_{i,t})\}_{t=1}^{T_i}$, I show how firms' decisions on employment adjustment (both on the extensive and intensive margins) and exit are qualitatively and quantitatively affected by those.

2.2 Log-linear approximation of the employment policy function

Consider a log-linear approximation of the employment policy function, $n_{i,t} \equiv \varphi(n_{i,t-1}, \xi_{i,t})$:

$$\log(n_{i,t}) = \lambda_1 \log(n_{i,t-1}) + \lambda_2 \log(\xi_{i,t}) + \epsilon_{i,t} \tag{2}$$

While λ_1 measures the serial correlation of log employment, λ_2 gauges the responsiveness of labour demand to contemporaneous profitability innovations: $\lambda_2 = \partial \log(n_{i,t}) / \partial \log(\xi_{i,t})$. Table 2 shows their estimates.

Table 2: Log linear approximation of employment policy function

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Serial correlation, λ_1	0.953^{***}	0.949^{***}	0.960^{***}
	(0.005)	(0.006)	(0.004)
Responsiveness, λ_2	0.035^{***}	0.028^{***}	0.051^{***}
	(0.008)	(0.008)	(0.010)
Industry & Year FE	Yes	Yes	Yes
R^2	0.959	0.958	0.959
Observations	3 308 659	1 709 959	1 350 617

Notes: Standard errors are clustered at the industry level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

There are three results which stand out. First, the impressive quality of the log linear approximation. Roughly 96% of the variation in employment is explained with two single variables: lagged employment and current profitability. Second, the exceeding high values of λ_1 indicate employment remains an extremely persistent phenomenon all throughout the sample. Third, and more importantly, the estimated responsiveness is almost twice as large in the post-reform period (0.051) than it was before the intervention (0.028).

These results are robust to alternative specifications of the data generating process — see Appendix C. The third one, however, merits further consideration. Employment reacts positively to profitability innovations and that effect is almost twice as large in the period following the labour market reform. It is a result which builds on a very specific and structural view of the economy. Yet it is also mirrored in reduced-form objects. The covariance between log profitability and log employment, for example, rose from 0.64 in 2006-2013 to 0.81 in 2014-2019. For a more intuitive interpretation, the corresponding correlation coefficient had a statistically significant increase from 0.47 to 0.55.

Lastly, notice the dynamics of employment is the outcome of two broader decisions made by the firm: one at the *extensive* margin (adjustment/no adjustment), and another at the *intensive* margin (conditional on adjustment, size of job creation/destruction). These can be differently impacted depending on the formulation of labour adjustment costs. Take a reform which would essentially reduce a symmetric linear cost of adjustment. As the band of inaction is mostly driven by non-convex costs, the extensive margin decision would remain largely unchanged. But conditional on adjusting, the size of job growth would be different. Given the Portuguese reform targeted a reduction in fixed and linear costs alike, in the next sub-section I take a closer look at the extensive and intensive margins of adjustment.

2.3 Responsiveness regressions: employment adjustment decisions

Let the employment growth rate of firm i in period t be $\varkappa_{i,t} \equiv (n_{i,t} - n_{i,t-1})/\left(0.5 \times (n_{i,t} + n_{i,t-1})\right)$. The firm is assumed to be inside the band of inaction whenever the change in the number of workers is very small, *i.e.* smaller than 2.5% in absolute value, $\varkappa_{i,t} \in [-0.025, 0.025]$. The indicator variable $\mathbb{1}_{i,t}$ tracks this: let $\mathbb{1}_{i,t} = 1$ when the firm is outside that band of inaction, and 0 when it is inside.

Figure 3: Evolution of employment adjustments within band of inaction, $\varkappa_{i,t} \in [-0.025, 0.025]$

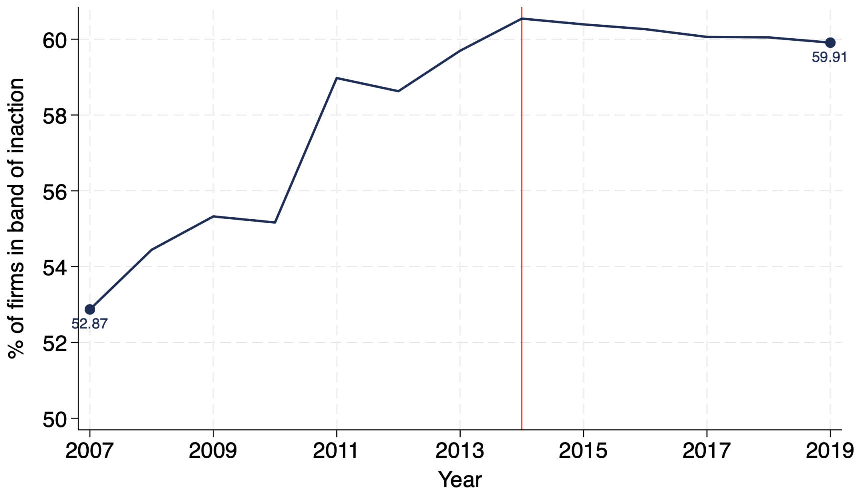


Figure 3 above shows the pervasiveness of employment inaction in the Portuguese economy. More than half of all yearly observations between 2007 and 2019 entailed employment adjustments smaller than 2.5% in absolute value [2]. This is a typical effect exerted by non-convex labour adjustment costs. Yet that inaction has been falling at a very slow pace since 2014: an average short of 0.15 percentage points each year. This is likely to be caused by the fact that new provisions introduced in the labour law were grandfathered in: although new hirings are covered by more flexible rules, pre-existing contracts are not. This duality can then act as a drag on inaction. I will return to this point below.

The responsiveness regression at the extensive margin corresponds to the following:

$$\mathbb{1}_{i,t} = \delta_1 \log(\xi_{i,t}) + \delta_2 \log(\xi_{i,t})^2 + \delta_3 \log(n_{i,t-1}) + \epsilon_{i,t} \tag{3}$$

Equation (3) posits how current profitability innovations and lagged employment determine the likelihood of an employment adjustment today. I use an identical formulation for responsiveness at the intensive

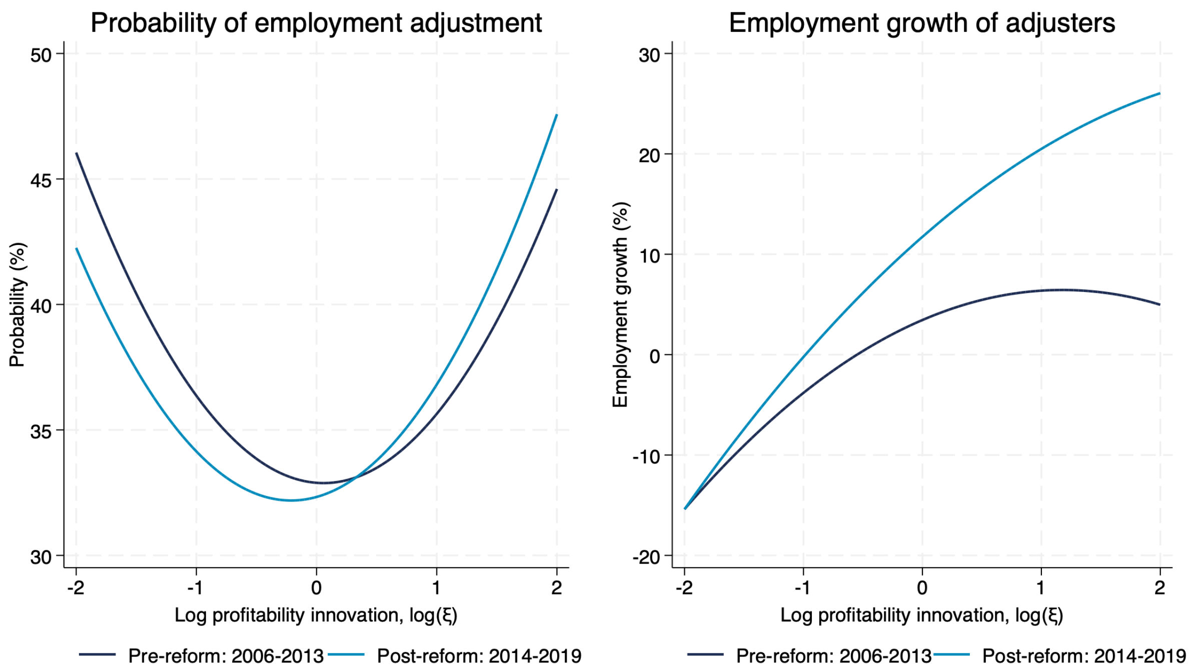
[2] For comparison, of the four European countries considered by Cooper et al. (2024b), Italy is the one where inaction is the greatest and only amounts to 35%.

margin. Conditional on action, the size of the employment adjustment is determined as follows:

$$\varkappa_{i,t\{\mathbb{1}_{i,t}=1\}} = \theta_0 + \theta_1 \log(\xi_{i,t}) + \theta_2 \log(\xi_{i,t})^2 + \theta_3 \log(n_{i,t-1}) + \epsilon_{i,t} \tag{4}$$

The results are reported in tables C7 and C8. I rely on those to project the left-hand side of both (3) and (4) against a very fine grid of log profitability innovations[3]. These are shown in Figure 4 below.

Figure 4: Responsiveness regressions at the extensive (left) and intensive (right) margins of employment



Notes: Extensive and intensive margin decisions of the average-sized firm. For the post-reform period of 2014-2019 size is kept constant at the 2006-2013 level.

The left panel on the probability of employment adjustment yields two major results. First, the LPM predicts a convex adjustment hazard: firms are more likely to adjust employment when hit by profitability innovations which are increasingly larger in absolute value. And second, there is an interesting asymmetry in the U-shaped curves. Conditional on drawing a *negative* profitability innovation, firms were more likely to respond to it prior to the reform; but when faced with a *positive* innovation, the likelihood of adjustment is greater in the post-reform period. This again appeals to the duality introduced by the reform. Although the new provisions were mostly aimed at firing costs, their immediate effect implied greater incentives for job creation. Quantitatively, a positive one standard deviation profitability innovation would raise the probability of adjustment by 0.9 percentage points prior to the reform; after the set of more flexible rules were put in

[3]The log profitability innovations are normally distributed with mean 0 and standard deviation close to 0.6 — see Table 1. The range $[-2, 2]$ in that grid encompasses 99.91% of the possible realisations.

place, an identical shock would actually raise the same probability by 1.9 percentage points.

The increase in responsiveness can also be found at the intensive margin. The right panel shows that, after the reform, employment adjustments carried out by firms do respond more to profitability innovations. The fact that the hiring rule is concave is consistent with the results documented by Ilut et al. (2018). But as costs of adjustment loosen, the hiring rule becomes less concave and increasingly linear. This further amplifies the response to both positive and negative one standard deviation profitability shocks. For firms which do adjust, and conditional on size, one such positive shock would raise employment growth by 2.3 percentage points before the reform, and by 5.6 afterwards. In case of a negative innovation, employment growth would fall by 3.8 percentage points before the reform, and by 6.7 afterwards.

2.4 Responsiveness regression: exit decision

Another extensive margin decision of firms concerns exit. This also relates to the existence of labour adjustment costs. Take a firm which is hit by a very negative profitability innovation. In the absence of labour frictions the firm would be able to costlessly downsize to its new optimal scale. Even accounting for potential fixed costs of operation, some firms would decide stay in the economy. But if the costs of firing are prohibitively high, that labour adjustment may not take place and the firm decides to exit altogether.

To study this decision let $\chi_{i,t}$ be an indicator variable which takes the value 1 when firm i exits the economy in the beginning of period t. The responsiveness to profitability can then be assessed with the following regression:

$$\chi_{i,t} = \psi_1 \log(n_{i,t-1}) + \psi_2 \log(\xi_{i,t-1}) + \epsilon_{i,t} \tag{5}$$

The timing of the independent variables is slightly different from that used in equations (3) and (4). A firm which exits in period t can only be observed in the data up to period $t-1$. This implies that the exit decision responds to the stock of workers and the profitability innovation drawn in that previous period. Table 3 below shows the estimated coefficients.

Two results stand out. First, the probability of exiting depends negatively on size. The more workers a firm has, the less likely it is to exit in the next period. This can be intuitively related to the concept of severance payments: an exit cost which grows linearly with the size of the firm. And second, those same exit decisions also depend negatively on profitability innovations. This is an expected result as firms which draw increasingly positive shocks are less likely to exit. However, that coefficient is not statistically different across the two time windows. This means that, conditional on size, the probability of exiting remains unaltered despite the documented changes in adjustment costs and the curvature of the revenue function.

Table 3: Log linear approximation of exit policy function

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Lag employment, ψ_1	-0.026^{***}	-0.028^{***}	-0.023^{***}
	(0.004)	(0.004)	(0.004)
Responsiveness, ψ_2	-0.073^{***}	-0.075^{***}	-0.072^{***}
	(0.010)	(0.010)	(0.010)
Industry & Year FE	Yes	Yes	Yes
Observations	3 308 659	1 709 959	1 350 617
R^2	0.116	0.125	0.106

Notes: Standard errors are clustered at the industry level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

2.5 Changes in the curvature of the revenue function

The preceding sub-sections have demonstrated that the employment responsiveness of Portuguese firms rose once labour adjustment costs were loosened. But contrary to what was ultimately intended, the distribution of labour productivity remained unaltered. The culprit lies with the result reported in Table 1. However more responsive firms may be, a reduction in the curvature of the revenue function acts as a break on labour demand adjustments: since the marginal worker does not bring as much revenue, enhanced responsiveness does not translate into productivity gains. Given the importance of this mechanism to the main message of the paper, in this sub-section I provide tentative evidence and discuss multiple economic forces which can explain the estimated fall in the curvature of the revenue function.

Say the *true* data generating process of the economy is exactly as described in sub-section 2.1. If so, then only *i)* changes in the output elasticity of labour, and/or *ii)* changes in the aggregate markup embedded in the price elasticity of demand can account for a dwindling curvature. Under some disputable assumptions, De Loecker and Syverson (2021) indicate that output elasticity can be read off standard firm-level data as the ratio between the wage bill and revenues, *i.e.* the labour share. Together with the GMM-IV estimates of α documented in Table 1, these values are sufficient to infer the price elasticity of demand and the associated aggregate markup. This exercise is carried out in Table 4 below.

The labour share averaged 0.29 in 2006-2013 and 0.32 in 2014-2019. With such negligible change most of the variation in the curvature must have actually come from the price elasticity of demand. This is corroborated by the counterfactual exercises. If that elasticity had remained unchanged and only the labour share had been allowed to vary, the counterfactual curvature would have been more than 33% greater than the observed value. As the isoleastic demand schedule becomes more inelastic, firms can charge higher markups.

Table 4: Disentangled changes in the estimated curvature of the revenue function

	Pre-reform: 2006-2013	**Post-reform: 2014-2019**
Curvature, α	0.563	0.442
Output elasticity of labour, ζ	0.285	0.316
Implied elasticity of demand, $\nu \equiv \|(1-\varepsilon)^{-1}\|$	5.527	3.511
Implied markup, $\mu \equiv \varepsilon^{-1}$	1.221	1.398
Counterfactual α: constant labour elasticity		-5.734%
Counterfactual α: constant markup		$+33.05\%$

Notes: The curvature of the revenue function corresponds to $\alpha \equiv \frac{\zeta\varepsilon}{1-\varepsilon(1-\zeta)}$. The labour share of a firm is computed as the ratio between the wage bill (including social security contributions) and its revenues. The output elasticity of labour, ζ, is obtained by taking the unweighted mean of (1% winsorized) firm-level values and then the time-series average across the corresponding time window. The number of observations in each time window is the same as reported in Table 1.

Can the reduction in the curvature of the revenue function be explained by rising product market power? In Table B4 I resort to the method of De Loecker and Warzynski (2012) and show there is indeed a rightward shift in the distribution of firm-level markups. This would also be consistent with a finding reported by Biondi et al. (2026): on average, markups rose in the set of European countries which experienced a decline in employment responsiveness. According to De Loecker et al. (2020) and Baqaee and Farhi (2020), this increase in product market power can have a detrimental impact on allocative efficiency which ultimately explains the muted response of productivity.

But plausible as this explanation may be, it also rests on a set of very strict assumptions. For example, besides a constant returns to scale technology, the recovery of the output elasticity of labour as an average of firm-level labour shares implies that labour market wedges produced by adjustment costs be mean-zero across the two periods. Since the reform is likely to have altered the structure of wedges in the labour market, the conclusion of an increasing aggregate markup follows from biased estimates of ζ. Just as importantly, that same result builds on a framework which assumes an isolastic demand schedule: $\log(y) = \log(\theta) - \nu \log(p)$. This is not compatible with the variable firm-level markups reported in Table B4. It is also inconsistent with the estimated non-linearities in the demand functions reported in Santos et al. (2022) for Portuguese manufacturing industries.

Therefore, a smaller curvature may actually mirror broader changes in the demand structure of the economy which are unrelated to changes in markups. Suppose now the reported revenue of firms is an aggregation of revenue channeled from multiple markets. For example, it can reflect sales of more than just one product, and/or exporting activities. If the elasticity of substitution between those different markets changes, that will inevitably be picked up by the estimates of α. Although a more careful investigation would require that I take a particular stand on a richer demand system, here I just provide two pieces of evidence which suggest such phenomena beyond rising product market power might be at play. First, even though more than half of all Portuguese manufacturing businesses still produce more than one product[4], Figure B1 reveals there has been a steady increase in the share of single-product firms. And second, while employment shares of the five largest sectors in the economy have barely changed, their export shares have become more pronounced — see Table B5. Neither of these developments appeals to rising product market power exerted by firms. Yet they are both consistent with a change in the effective demand elasticity faced by firms, which is ultimately reflected in the estimated values of the curvature.

It is not the purpose of this paper to adjudicate amongst all these alternative explanations. It is also not part of its scope to ascertain whether the change in the curvature is an *endogenous* development of the reform or a coincidental *exogenous* phenomenon. The distinction does not taint one of the main contributions of the paper: employment responsiveness increased following a labour market reform which slashed adjustment costs in the Portuguese economy. It is, however, a relevant difference to determine if similar policies can be generally rendered ineffective tools to boost productivity. I will revisit this point in section 5.

In conclusion, the fall in the curvature of the revenue function is to be taken as a documented fact whose underlying source is yet to be properly explained. But it is an interesting fact on its own: a reduced curvature mutes the productivity effects generated by a rising employment responsiveness. To corroborate this claim in the following sections I develop and estimate and partial equilibrium model of labour demand. Its goal is twofold. First, it provides a set-up which can be relied upon to rationalise the observed changes in employment responsiveness and labour productivity. And second, it allows to formally express moments of the labour productivity distribution as a function of three key mechanisms: *i)* labour adjustment costs, *ii)* the curvature of the revenue function, and *iii)* the stochastic process of profitability. While Table 1 provides estimates for the latter two, the adjustment cost function is estimated via simulated method of moments.

3 Model economy

I build a partial equilibrium model of dynamic labour demand with adjustment costs. The unit of analysis is the firm, which also makes endogenous decisions about entering and exiting the economy. The framework follows closely those of Hopenhayn (1992) and Hopenhayn and Rogerson (1993).

At the beginning of the period surviving firms draw their idiosyncratic profitability level, $A \in \mathcal{A} \subset \mathbb{R}_+$, and know the stock of workers they inherit from the previous period, $n_{-1} \in \mathbb{R}_+$. The stationary AR(1)

[4]The underlying data are provided by an yearly survey on Portuguese manufacturing firms (Statistics Portugal (2025)).

process which governs A is that specified in Section 2. Firms immediately rely on this duple, (A, n_{-1}), to determine if they want to remain in the economy as incumbents or to exit permanently. Formally:

$$V(A, n_{-1}) = \max \left\{ \underbrace{V^i(A, n_{-1})}_{\text{Incumbents}}, \underbrace{V^o(n_{-1})}_{\text{Exiters}} \right\} \tag{6}$$

Incumbents:

Firms which decide to continue their operations use (A, n_{-1}) to optimally choose employment (number of workers to be employed instantly (n), and hours per worker (h). Formally, their dynamic programming problem is expressed as follows:

$$V^i(A, n_{-1}) = \max_{n \geq 0, h \geq 0} \left\{ R(A, n, h) - \omega(n, h) - c - C(n_{-1}, n) + \beta \int_{A' \in \mathcal{A}} V(A', n) \cdot \mathrm{d}\Pi(A'|A) \right\} \tag{7}$$

The flow of revenue to the firm is determined by the profitability level and the labour demand choices: $R(A, n, h) \equiv A \cdot (nh)^\alpha$. The chosen combination of n and h requires, however, that compensation be paid in the amount of $\omega(n, h) \equiv n \cdot (w_0 + w_1 h^\eta)$, where w_0 is the base wage, w_1 the hourly rate, and $\eta > 1$ the constant elasticity between the hourly wage cost and hours per worker[5]. This formulation assumes that the decision on average hours is static: firms merely trade off the marginal revenue associated with raising average hours of work with the contemporaneous marginal wage cost of doing so.

The choice of the number of workers, on the other hand, is dynamic as it is subject to an adjustment cost function, $C(n_{-1}, n)$. This function has a piecewise formulation to create the distinction between *net* costs of job creation and destruction, and nests both fixed and linear forms of adjustment.

$$C(n_{-1}, n) \equiv \begin{cases} \Gamma_f + \gamma_f \cdot (n_{-1} - n) & \text{if } n_{-1} > n \\ 0 & \text{if } n_{-1} = n \\ \Gamma_h + \gamma_h \cdot (n - n_{-1}) & \text{if } n_{-1} < n \end{cases}$$

The inclusion of non-convexities, Γ_f and Γ_h, is meant to capture the pervasiveness of inaction first reported by Varejão and Portugal (2007) and also documented in Section 3 before. These can also encompass measures such as the introduction of a more flexible reason to justify fair dismissals. On the other hand, the linear components, γ_f and γ_h, can capture changes to severance payments and searching costs. The subscripts f and h denote *firing* and *hiring*, respectively.

Lastly, incumbents must also bear a fixed cost of production, c. The existence of such a cost generates endogenous exit, which is taken to be permanent.

[5]The convexity of the hourly wage schedule is justified with the data. Hours and lagged employment covary negatively: the correlation is -0.119 in the whole sample. The optimal decision on hours implies that $\frac{\partial h}{\partial n_{-1}} = \frac{\alpha-1}{\eta-\alpha} \cdot \varphi(A, n_{-1})^{\frac{\alpha-1}{\eta-\alpha}-1} \cdot \varphi_2(A, n_{-1})$. Under $\varphi_2(A, n_{-1}) > 0$, as estimated in sub-section 2.2, $\eta > 1 > \alpha$ ensures the existence of that negative relationship.

Exit and entry:

Exit is not free. Firms which find it optimal to permanently shut down their operations must pay severance to all workers who are on payroll. Since no other liabilities are due, the value of exiting is the following:

$$V^o(n_{-1}) = -\gamma_f \times n_{-1} \tag{8}$$

The model also includes endogenous entrance in the economy. For each exiting firm there exists a potential entrant which is endowed with a signal about their prospective profitability in the next period. That signal, $\tilde{A} \in \mathcal{A}$, is drawn from the stationary distribution. The firm can then decide to enter the following period and start operating at the lowest level of employment, $\underline{n} \in \mathbb{R}_+$, if and only if the conditional expected value of becoming an incumbent is weakly positive:

$$\int_{A' \in \mathcal{A}} V(A', \underline{n}) \cdot \mathrm{d}\Pi(A'|\tilde{A}) \geq 0 \tag{9}$$

Stationary equilibrium:

Absent aggregate shocks, a stationary equilibrium of the economy corresponds to a steady-state distribution over (A, n). That distribution must exist and is defined by the following elements:

1. The profitability conditional CDF, $\Pi(\cdot|A)$;
2. The employment policy function, $n \equiv \varphi(A, n_{-1})$, which solves the dynamic problem in (7);
3. The exit policy function, $\chi(A, n_{-1})$, which takes the value 1 when $V^o(n_{-1}) > V^i(A, n_{-1})$, and 0 otherwise;
4. The entrance policy function, $\psi(\tilde{A})$, which takes the value 1 when condition (9) holds, and 0 otherwise.

4 Estimation

The model includes a revenue function whose parameters α, ρ, and σ were estimated in sub-section 2.1. There are seven remaining parameters to identify: $\Theta \equiv (\gamma_f, \Gamma_f, \gamma_h, \Gamma_h, \eta, w_0, c) \in \mathbb{R}^7$. In this section I describe their estimation procedure, the results, and discuss the strength of the identification scheme.

4.1 Estimation approach

I rely on a simulated method of moments (SMM) to estimate Θ. The corresponding estimator, $\hat{\Theta}$, minimises the following objective function, F:

$$\hat{\Theta} \equiv \arg\min_{\Theta \in \mathbb{R}^7} \left\{ \left(\left(m^d - m^s(\Theta) \right) / m^d \right) W \left(\left(m^d - m^s(\Theta) \right) / m^d \right)' \right\}, \tag{10}$$

where $m^s(\Theta)$ is the vector of simulated moments obtained when the model is parameterised by Θ, and m^d the data counterparts of those moments. Below I describe the seven targeted moments and offer some preliminary justification as to why they were picked. Since the model is just-identified, the weighting matrix, W, is the conforming identity matrix.

Identification:

Identification concerns the choice of targeted moments. These are listed in Table 5.

Table 5: Targeted moments in structural estimation

ϱ_{An}	Covariance b/w TFPR and employment	λ_1	Serial correlation log employment
θ_2	Concavity of hiring rule	ψ_2	Responsiveness of exit decision
$\sqrt{\Sigma_{\overline{r}}}$	Std deviation of log labour productivity	$\overline{n}$	Mean of log employment
$\overline{x}$	Time-series average of exit rate		

I now focus more on the covariance between the log profitability and log employment, ϱ_{An}. This is justified by two reasons. First, because this is an object with an intrinsic connection to concepts of responsiveness and adjustment costs. If adjustments are prohibitively expensive, for example, then both the covariance and responsiveness approach zero because firms do not alter their labour force however profitable they may be. And second, because it will be shown to be important to decompose the variance of labour productivity.

The following three moments correspond to coefficients from regressions explored in the previous section. The effect exerted by non-convexities, for instance, is well captured by the serial correlation of log employment in the policy function (2), λ_1. Although those non-convexities also impact the responsiveness of the exit decision, the latter is highly influenced by the size of linear costs of firing. The coefficient ψ_2 in (5) can be quite important to identify those. Reduced severance payments slash the cost of exiting and this may prompt firms to respond differently when hit by a negative realisation of profitability. This notwithstanding, the existence of convex costs is best mirrored in the intensive margin decisions of employment adjustment. As a consequence, I also rely on θ_2, the coefficient which defines the concavity of the hiring rule in (4).

The remaining moments are usual in the literature. Decisions on hours are tamed by two components of the wage function: the hourly rate, w_1, and the elasticity between the hourly wage cost and hours per worker, η. I keep the hourly rate outside the estimation procedure and calibrate it such that the steady-state value of hours per worker matches the yearly average found in the data. The elasticity, on the other hand, is kept within the estimand vector and is mostly targeted by the standard deviation of labour productivity, $\sqrt{\Sigma_{\overline{r}}}$. Finally, the mean of log employment, $\overline{n}$, is highly influenced by the base wage (w_0), while the average exit rate, $\overline{x}$, is exceptionally informative about the fixed cost of operation (c).

4.2 Estimation results

Parameters:

I estimate the model parameters for two time windows: one targeting the data moments of 2006-2013, and another one targeting those of 2014-2019. In doing so I can capture the effects the labour market intervention has had on the labour adjustment cost parameters and therefore on the various moments. Table 6 below shows those point estimates in the two periods.

Table 6: Parameter estimates

	Labour adjustment costs				Wage function		Fixed cost
	γ_f	Γ_f	γ_h	Γ_h	η	w_0	c
2006-2013	0.23	0.55	0.18	0.13	4.25	1.25	0.04
2014-2019	0.19	0.35	0.27	0.07	4.16	0.73	0.06

Notes: All labour adjustment costs parameters and the fixed cost of operation are expressed as a share of the average revenue of the simulated economy. The economy is simulated at a quarterly frequency and aggregated to match the annual moments of the data. The quarterly discount factor is set to $\beta = 0.99$, and the hourly rate w_1 is calibrated such that the steady-state value of hours per worker replicates the observed yearly average.

These estimated parameters are consistent with a sizeable cut in firing costs. Between 2006 and 2013, a firm was predicted to forego 78% of the economy's average revenue when sacking a single employee. This entails both the fixed cost and a linear component reminiscent of severance payments. That cost was brought down to 54% in the subsequent period. Evidence on hiring costs is not as clear. Taking one additional employee would originally wipe out 31% of the average revenue. Now, it does so by 34%.

The elasticity between average hours of work and the hourly wage cost, η, remains largely unchanged. For a 1% increase in average hours, the hourly wage bill of firms was estimated to increase by 4.25% in the first time window, and by 4.16% in the second one. Finally, the fixed cost of operation is estimated to increase from 4% to 6% of the average revenue.

Targeted moments:

The fitness of the SMM estimation is reported in Table 7 below. Each of the two blocks shows the data and simulated moments in their corresponding time window.

The sum of squared distances between data and simulated moments, $F(\hat{\Theta})$, is noticeably small in the two periods. This indicates the model does a successful job replicating the targeted moments. This is particularly true for the responsiveness moments, both qualitatively and quantitatively. For example, the model is able to reproduce the observed concavity of the employment hiring rule, $\theta_2 < 0$; it also picks up the significant

Table 7: SMM estimation fit — targeted moments

		Responsiveness				Distributions		Exit	Distance
		ϱ_{An}	λ_1	θ_2	ψ_2	$\sqrt{\Sigma_{\overline{r}}}$	$\overline{n}$	$\overline{x}$	$F(\hat{\Theta})$
2006-2013	*Data*	0.64	0.95	-0.02	-0.08	1.14	1.24	8.16	
	Model	0.58	0.83	-0.02	-0.07	1.77	1.17	6.03	0.41
2014-2019	*Data*	0.81	0.96	-0.02	-0.07	1.16	1.14	6.44	
	Model	0.67	0.83	-0.02	-0.08	1.59	1.16	6.19	0.21

Notes: Targeted moments are the covariance between the log level of TFPR and log employment ϱ_{An}, the serial correlation of log employment λ_1 in (2), the concavity of the hiring rule θ_2 in (3), the exit responsiveness coefficient ψ_2 in (5), the standard deviation of log labour productivity $\sqrt{\Sigma_{\overline{r}}}$, the mean of log employment $\overline{n}$, and the mean of the exit rate $\overline{x}$. The measure of distance between data and simulated moments defined in (10) is $F(\hat{\Theta})$.

increase in the covariance between log TFPR and log employment. Although not shown here, this model-based increase in responsiveness can also be found in untargeted moments such as the parameter in the policy function for labour, λ_2, and the correlation coefficient between log profitability and log employment.

The predicted values of the *standard deviation* of log labour productivity are not substantially off from their data counterparts: 1.77 vs. 1.14 in the pre-reform period, 1.59 vs. 1.16 afterwards. Yet, however small these differences may be, they compound once I rely on the *variance*. This moment is then projected to be 3.14 in the 2006-2013 period (1.31 in the data), and 2.53 in 2014-2019 (1.35 in the data)[6]. Conversely, both the exit rate and specially the mean of the employment distribution are very well targeted. In Table D3 I show the model can also replicate the flat evolution of the cross-sectional mean of labour productivity.

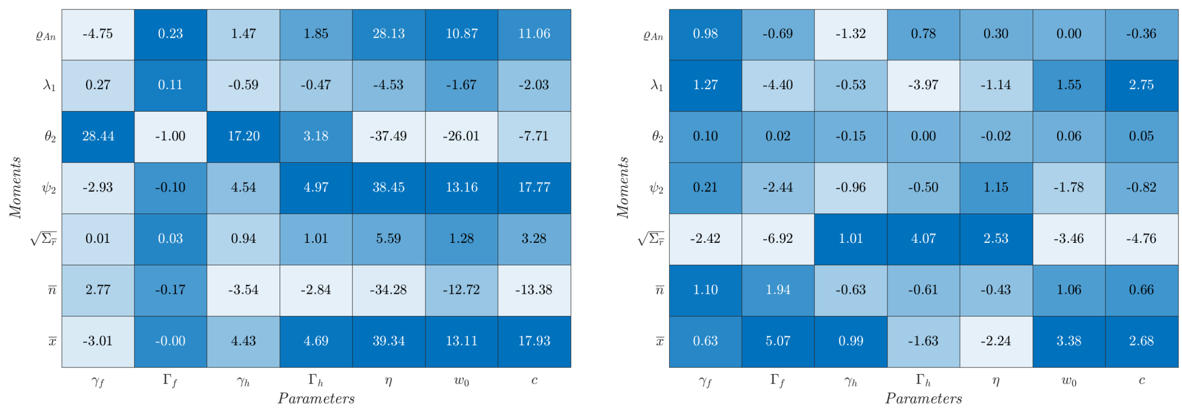
Robustness of the identification scheme:

The strength of the identification scheme is assessed in two different ways. In Figure 5 below I uncover the intricate mapping from parameters to moments. More specifically, elements in left panel indicate the percentage change in each targeted moment following a 1% change in each parameter, all else equal. In the right panel I rely on the methodology of Andrews et al. (2017) and implement the reverse exercise: how much do parameters vary following changes in the targeted moments.

The linear cost of firing, reminiscent of severance payments, is particularly well identified. All else equal, raising those linear costs by 1% reduces the covariance between log TFPR and log employment by 4.75%,

[6] An alternative option would then be to immediately target the variance instead of the standard deviation. Yet the overall performance of the model is considerably worse in that case. To reduce the dispersion of productivity the minimisation algorithm raises adjustment costs and η to the point where the economy is only populated by very large firms with similar labour productivity levels. Since there is hardly any exit, the new solution gets the variance almost right but at the expense of the two moments connected to that decision, ψ_2 and $\overline{x}$.

Figure 5: Local elasticities of moments wrt parameters (left) and parameters wrt moments (right)



Notes: In the *left panel* the element (m, p) of the matrix gives the local elasticity of moment m following a 1% change in parameter p. In the *right panel* element (m, p) of the matrix gives the local elasticity of parameter p following a 1% change in moment m. All these local elasticities are computed when the parameters are identified for the full period of the sample, 2006-2019.

increases the average size of firms by 2.77% and, as a consequence, the exit rate drops by 3.01%. These are meaningful results: severance payments took centre stage in the reform and they are shown to exert widespread and sizeable effects in the economy.

The dispersion of labour productivity, on the other hand, is immune to them. Although the standard deviation does respond almost on a one-to-one fashion to linear and fixed costs of hiring, it is mostly sensitive to changes in the elasticity between the hourly wage cost and hours per worker, and the fixed cost of operation. And as argued before, the average exit rate is mostly informed by the latter.

Alternative parameterisation:

The estimation algorithm described in equation (10) aimed to minimise a sum of equally-weighted squared distances between simulated and data moments. I now consider an alternative estimation strategy. More specifically, I follow Adda and Cooper (2003) and rely on a two-stage SMM procedure which takes advantage of the optimal weighting matrix. Conditioning on the baseline estimates reported in Table 6, I simulate 1 000 economies and store each of the resulting vectors of simulated moments. The inverted variance-covariance matrix of all those moments is then plugged back in (10) to take the role of weighting matrix in a new iteration of the minimisation procedure.

The point estimates can be found in Table D1, whereas the comparison between data and simulated moments is shown in Table D2. Overall, the results are consistent with those discussed here. If anything, the reduction in labour adjustment costs is made slightly more prominent. Sacking one employee would cost up to 100% of the average revenue between 2006 and 2013, but only 64% in the subsequent period. The evidence on unit costs of hiring, on the other hand, remains less clear. Hiring one more employee would initially cost

35% of the average revenue; it now takes 39%.

5 Quantitative analysis

In this section I rely on the estimated version of the model to shed light on the muted evolution of the labour productivity variance, a measure tightly linked to the concept of misallocation.

I start with a counterfactual exercises which confirm that, albeit simple, the estimated model encompasses the necessary mechanisms to better understand the documented facts regarding not just labour productivity but also employment responsiveness. While keeping everything else constant, I separately alter the parameter values of the labour adjustment cost function, and the curvature of the revenue function. The goal is to disentangle the effects each channel produces. In sub-section 2.5 I raised the possibility these two mechanisms may be endogenously linked. For example, with cheaper exit and looser adjustment constraints, exceedingly profitable firms may grow disproportionately more and can thus consolidate a more dominant position. Lower costs of adjustment would raise market power, which would then be mirrored in the curvature of the revenue function. If these two channels are not independent, an understandable concern is whether these counterfactuals have a clean policy interpretation. Suppose indeed they are not. The purpose of this analysis is to confirm the observed evolution of labour productivity is the outcome of two offsetting channels. Even if the reform jointly determines labour adjustment costs and the curvature of the revenue functions, there is no reason to expect their individual effects on labour productivity should be quantitatively similar. The potential presence of an endogenous link does not distort the validity of the exercise. It does, however, cast doubt on the general ability of an increasingly flexible labour market to function as a catalyst of productivity gains.

In a second exercise I emphasise the impact of the labour market reform alone. To do so I keep all other parameters constant at their 2006-2013 values and just re-estimate the components of $C(n_{-1}, n)$ while targeting responsiveness moments and the dispersion of labour productivity in 2014-2019. This partial re-estimation exercise aims to show that both responsiveness and labour productivity are indeed sensitive to changes in costs of adjustment.

In the last subsection I demonstrate that the variance of labour productivity can ultimately be written as a closed-form expression of the curvature of the revenue function and the variance-covariance matrix of TFPR, employment, and hours worked. Since the cross-sectional dispersion of TFPR has long been interpreted as a measure of misallocation, studying the decomposition of the labour productivity variance is of unequivocal importance for macroeconomists. I pinpoint which elements are more prominent to explain that dispersion in the data, in the baseline version of the model, and in counterfactual economies where labour adjustment costs and the whole profitability process of firms had not changed.

5.1 Counterfactuals

I consider three types of counterfactual cases. In the first one I ask what would happen to the economy if, all else equal, the labour market reform would be fully reversed. Call this scenario "**Counterfactual,** $C(n_{-1}, n)$". To be clear, this means that while the four elements which describe the labour adjustment cost function $C(n_{-1}, n)$ retake their initial values, all other parameters of the economy are kept at their 2014-2019 levels. This includes the curvature of the revenue function, the serial correlation and variance which describe the profitability process, the components of the wage function, and the fixed cost of operation. It is by doing so that I can precisely identify the effects exerted by labour adjustment costs alone. In the second one I report the effects which would have been produced had policy-makers gone farther and removed all sorts of adjustment costs. Call this case "**Counterfactual**, *free adjustments*". While the first exercise describes a movement towards a more stringent labour market, the second one implies a full liberalisation. Lastly, in the third case I study a fictional economy where all parameters would keep their most recent values with the exception of the curvature of the revenue function. Let this scenario be called "**Counterfactual,** α".

Table 8: Counterfactual exercises

		Targeted moments		Distance
		ϱ_{An}	$\sqrt{\Sigma_{\bar{r}}}$	$F(\hat{\Theta})$
2006-2013	*Data*	0.64	1.14	
	Model	0.58	1.77	0.41
2014-2019	*Data*	0.81	1.16	
	Model	0.67	1.59	0.21
	Counterfactual, $C(n_{-1}, n)$	0.31	1.76	1.53
	Counterfactual, *free adjustments*	2.79	1.03	19.08
	Counterfactual, α	0.24	1.36	8.95

Notes: In **Counterfactual,** $C(n_{-1}, n)$ only the four adjustment cost parameters ($\gamma_f, \Gamma_f, \gamma_h$, and Γ_h) are kept at their 2006-2013 levels. All other parameters take their 2014-2019 values.

In **Counterfactual,** *free adjustments* all four adjustment costs parameters ($\gamma_f, \Gamma_f, \gamma_h$, and Γ_h) are set to zero. All other parameters take their 2014-2019 values.

In **Counterfactual,** α only the curvature of the revenue function (α) is kept at its 2006-2013 levels. All other parameters take their 2014-2019 values.

Table 8 shows the outcome of these exercises. I focus on two moments which were targeted in the estimation procedure: *i)* the covariance between profitability and employment, and *ii)* the standard deviation of log labour productivity. The distance between the whole set of data and simulated moments is reported in the last column. In Table D3 I replicate this exercise with a set of untargeted moments.

Start with the effects produced by labour adjustment costs only. The first two exercises confirm that, all else equal, a more rigid/flexible labour market is associated with falling/rising responsiveness but rising/falling labour productivity dispersion. If the labour market reform would be reversed, the covariance would be halved (from 0.67 to 0.31) and the dispersion of labour productivity would increase 11% (from 1.59 to 1.76). Conversely, if policy-makers had completely wiped out all adjustment costs, the covariance would have increased by a factor of four, and the standard deviation would have dropped 35%.

Consider now the counterfactual economy where, all else equal, the curvature of the revenue function would return to its original higher value. The dispersion of labour productivity would be smaller, as the standard deviation would fall from 1.59 to 1.36. This means that had it not been for a lower curvature of the revenue function, that dispersion would have fallen 14%. Recall that reversing the labour market reform would have increased it 11%. In absolute terms, these are remarkably similar effects, which suggests that the two channels do exert a quantitatively neutral impact on the labour productivity dispersion. This explains why the documented increase in responsiveness was not followed by noticeable changes in the variance of labour productivity: as the curvature of the revenue function fell, it fully offset the positive effects lower costs of adjustment would bring.

Lastly, notice that the fit of the model worsens in all three counterfactual scenarios relative to the baseline. Yet the case where the change is the smallest is the one where I simulate a full reversal of the labour market reform. This could potentially be construed as evidence that the set of targeted moments is not very sensitive to changes in the parameters of the adjustment cost function. To discredit this hypothesis, in the next subsection I demonstrate that appealing solely to changes in labour adjustment costs is sufficient to target most responsiveness coefficients and the standard deviation of labour productivity.

5.2 Re-estimation of labour adjustment costs

In the previous exercise I separately switched off the two mechanisms deemed responsible for the change in responsiveness and the standard deviation of labour productivity. I now do something slightly different. With the exception of the four parameters which describe the labour adjustment cost function, suppose all others would have been kept constant at their 2006-2013 levels. Again, this includes the curvature of the revenue function, the serial correlation and variance of the profitability process, the components of the wage function, and the fixed cost of operation. In order to focus more explicitly on the effects brought about by a more flexible labour market, I re-estimate the four parameters which fully describe $C(n_{-1}, n)$ while targeting *i)* the covariance between profitability and employment, *ii)* the serial correlation of log employment, *iii)* the concavity of the hiring rule, and *iv)* the standard deviation of log labour productivity. This just-identified estimation exercise aims to reinforce how adjustment costs alone can exert a significant influence in both employment responsiveness and the dispersion of labour productivity. The results are displayed in Table 9.

Table 9: Partial re-estimation of labour adjustment costs

		Targeted moments				Distance
		ϱ_{An}	λ_1	ψ_2	$\sqrt{\Sigma_{\bar{r}}}$	$F(\hat{\Theta})$
2006-2013	*Data*	0.64	0.95	-0.08	1.14	
	Model	0.58	0.83	-0.07	1.77	0.41
2014-2019	*Data*	0.81	0.96	-0.07	1.16	
	Model	0.67	0.83	-0.08	1.59	0.21
	Partial re-estimation	0.83	0.84	-0.07	1.30	0.03

Notes: In **Partial re-estimation** all but the four adjustment cost parameters are kept at their 2006-2013 levels. These are estimated while targeting the four moments displayed in the table: ϱ_{An}, λ_1, ψ_2, and $\sqrt{\Sigma_{\bar{r}}}$.

The distance between data and simulated moments is only 0.03, a remarkably smaller value than in the baseline (0.21). This indicates the model does a great job replicating the targeted moments. Put differently, by appealing solely to changes in labour adjustment costs one can reproduce various patterns observed in the data. The concavity of the hiring rule (-0.07) and the covariance between profitability and employment (0.83), for example, match almost exactly their data counterparts: -0.07 and 0.81, respectively. The model-predicted value for the standard deviation of labour productivity is also of interest. The closer the model can be to the observed value is actually in this partial re-estimation exercise.

5.3 Variance decomposition of labour productivity

In Tables 8 and 9 I assessed the macroeconomic implications of changing responsiveness mostly through the standard deviation of labour productivity. The latter is a measure intrinsically connected to the concept of misallocation. Surprisingly, despite all the significant transformations in the Portuguese economy, I showed in Figure 1 that standard deviation remained fairly stable between 2006-2013 and 2014-2019. But that does not mean that there cannot be substantial dynamism in the individual elements which jointly determine that dispersion. In this sub-section I start by deriving a closed form solution to the decomposition of that variance and then assess the relative contribution of all those elements both in the data and the estimated model. The usefulness of this exercise is twofold. First, it allows me to formally establish the connection between the dispersion of labour productivity and the misallocation measure defined in Hsieh and Klenow (2009): the cross-sectional variance of log TFPR. And second, it offers an adequate framework to study which economic forces are more important to understand the variation in labour productivity.

The natural logarithm of labour productivity corresponds to the difference between log revenues and log employment. Using the revenue function derived initially, and omitting firm- and time-specific subscripts for convenience of notation, I can express it as $\log(R) - \log(n) \equiv \log(A) + (\alpha - 1)\log(n) + \alpha\log(h)$. From the properties of the variance of a sum:

$$\mathrm{Var}\big(\log(R) - \log(n)\big) = \mathbf{tr}\left(\begin{bmatrix} 1 & 2(\alpha-1) & 2\alpha \\ 0 & (\alpha-1)^2 & 2\alpha(\alpha-1) \\ 0 & 0 & \alpha^2 \end{bmatrix} \cdot \begin{bmatrix} \mathrm{Var}\big(\log(A)\big) & \mathrm{Cov}\big(\log(n), \log(A)\big) & \mathrm{Cov}\big(\log(h), \log(A)\big) \\ \mathrm{Cov}\big(\log(A), \log(n)\big) & \mathrm{Var}\big(\log(n)\big) & \mathrm{Cov}\big(\log(h), \log(n)\big) \\ \mathrm{Cov}\big(\log(A), \log(h)\big) & \mathrm{Cov}\big(\log(n), \log(h)\big) & \mathrm{Var}\big(\log(h)\big) \end{bmatrix}\right) \tag{11}$$

Proof. See Appendix D. □

The variance of labour productivity is simplified to an explicit function of two elements: *i)* the curvature of the revenue function, which fully parameterises the first matrix, and *ii)* the variance-covariance matrix of profitability, employment, and average hours of work. Given these two matrices one just needs to compute the trace of the product between them.

For what follows I will partition the two matrices and just focus on the first 2×2 block matrices. This allows me to abstract from any variance or covariance elements associated with hours decisions. The discarded elements explain no more than 4% of the observed labour productivity variance. In other words, roughly 96% of the observed dispersion in labour productivity can be explained by the variance-covariance elements between profitability and employment, the focus of the paper.

$$\mathrm{Var}\big(\log(R) - \log(n)\big)^{-1} \times \underbrace{\mathbf{tr}\left(\begin{bmatrix} 1 & 2(\alpha-1) \\ 0 & (\alpha-1)^2 \end{bmatrix} \cdot \begin{bmatrix} \Sigma_A & \varrho_{An} \\ \varrho_{An} & \Sigma_n \end{bmatrix}\right)}_{=\Sigma_A + (\alpha-1)^2\Sigma_n + 2(\alpha-1)\,\varrho_{An}} \approx 0.96, \tag{12}$$

where Σ_A and Σ_n are the respective variances of log profitability and employment, and ϱ_{An} is the covariance between them. This last moment is one of the targets in the structural estimation exercise.

Decomposing the variance

The decomposition derived in (12) is useful to understand which economic forces more prominently explain the variance of labour productivity. In Table 10 I report the total variance, the level of these three elements which comprise it, and the sum of their corresponding shares.

Table 10: Variance decomposition of log labour productivity

		Variance	Elements of variance decomposition			Share of variance
		$\Sigma_{\bar{r}}$	Σ_A	$(\alpha-1)^2\,\Sigma_n$	$2(\alpha-1)\,\varrho_{An}$	
2006-2013	*Data*	1.31	1.60	0.23	-0.56	0.96
	Model	3.14	1.15	1.37	-0.51	0.64
2014-2019	*Data*	1.35	1.82	0.37	-0.90	0.96
	Model	2.53	1.34	1.17	-0.74	0.70
	Counterfactual, $C(n_{-1}, n)$	3.09	1.35	1.27	-0.34	0.74
	Counterfactual, $R(\cdot)$	1.86	1.58	0.04	-0.21	0.76

Notes: The first column displays the variance of log labour productivity, $\Sigma_{\bar{r}}$. The following three columns show the elements which are part of the variance decomposition: the variance of log profitability, Σ_A, the weighted variance of log employment, $(\alpha-1)^2\Sigma_n$, and the weighted covariance between log profitability and log employment, $2(\alpha-1)\varrho_{An}$. The last column reports the sum of the three preceeding elements divided by the variance of log labour productivity.

In **Counterfactual,** $C(n_{-1}, n)$ only the four adjustment cost parameters (γ_f, Γ_f, γ_h, and Γ_h) are kept at their 2006-2013 levels. All other parameters take their 2014-2019 values. In **Counterfactual,** $R(\cdot)$ all elements of the revenue function (α, ρ, and σ) are kept at their 2006-2013 levels. All other parameters take their 2014-2019 values.

Start with the decomposition found in the data. The variance of TFPR, Σ_A, plays a disproportionately large role: it corresponds to $(1.6/1.31) = 122\%$ and $(1.82/1.35) = 135\%$ of the labour productivity variance in the pre- and post-reform periods, respectively. The share of the weighted variance of employment, $(\alpha-1)^2\,\Sigma_n$, is smaller: only 17% and 28%. The varying shares mirror essentially changes in their numerators, *i.e.* changes in the variance of TFPR, and changes in the weighted variance of employment. The variance of profitability did increase from 1.6 to 1.82 and, more importantly, that rising dispersion can only be accounted for by the documented changes in ρ and σ because $\Sigma_A = \sigma^2/(1-\rho^2)$. The *unweighted* variance of employment remained constant, but the reported fall in the curvature of the revenue function reduced the *weighted* variance which is relevant to the decomposition.

These are two non-negligible evolutions which can be fully attributed to the estimated parameters of the revenue function. The model supports this claim. Suppose again the labour market reform would be reversed. Table 10 indicates that in the "**Counterfactual**, $C(n_{-1}, n)$" scenario both the variance of TFPR and the weighted variance of employment would be kept almost unchanged. This contrasts with a counterfactual exercise wherein all estimated parameters of the revenue function would retake their original values — call it "**Counterfactual,** $R(\cdot)$". Although the variance would drop and approach its data counterpart, there would be considerably more dynamism in its decomposition than that prompted by a tightening of labour adjustment costs.

6 Conclusion

This paper investigated whether a labour market reform which cuts adjustment costs can be an effective tool to boost productivity. In theory it should: looser constraints allow firms to be more responsive to changes in their profitability circumstances, and this improves the allocation of labour across the economy. However, a reform of this sort in Portugal did not fully deliver on its goal. Although labour adjustment costs were slashed, the declining trend in the pace of job flows across firms remained in place and productivity gains were nonexistent.

I start by addressing this puzzling result with a rich dataset of Portuguese firms. The conclusion drawn from an estimated log linear approximation of the employment policy function is unequivocal: the responsiveness of firms to exogenous profitability shocks almost doubled in the period after the labour market reform. This can be verified both at the extensive (a convex adjustment hazard) and intensive (a concave hiring rule) margins of labour adjustment. The likelihood of exit, however, was not affected.

As lawmakers cut labour adjustment costs, the employment responsiveness of Portuguese firms rose. This is a novel contribution to the literature. Yet that rising responsiveness has been incapable of speeding up the labour reallocation process from low- to more-profitable firms. The reason lies in the documented fall in the curvature of firms' revenue function. I corroborate this point with a partial equilibrium model of labour demand with adjustment costs, and endogenous entry and exit decisions. Through counterfactual exercises I demonstrate that, albeit in opposite directions, looser constraints in the labour law and a falling curvature exert quantitatively similar effects on the labour productivity variance, an object intrinsically linked to the concept of misallocation. In the data, a constant dispersion arises because these two channels offet each other.

This paper takes use of a very particular context to contribute to the debate on the sources of firms' employment responsiveness, and the implications for labour productivity. A promising line of research concerns the reasons underlying the fall in the curvature of the revenue function, a pivotal mechanism to understand the muted evolution of productivity. All the candidate explanations which were discussed merit deeper and more careful thought. I also did not ascertain whether that reduction should be interpreted as an exogenous standalone development or as an inevitable endogenous consequence. This distinction is not meaningless if policy-makers attempt to boost productivity through a more flexible labour market: the paper makes it clear the productivity gains from rising responsiveness would be wiped out as soon as the curvature of the revenue function would fall.

References

Adda, J. and Cooper, R. W. (2003). *Dynamic economics: quantitative methods and applications*. MIT press.

Andrews, I., Gentzkow, M., and Shapiro, J. M. (2017). Measuring the sensitivity of parameter estimates to estimation moments. *The Quarterly Journal of Economics*, 132(4):1553–1592.

Autor, D. H., Kerr, W. R., and Kugler, A. D. (2007). Does employment protection reduce productivity? evidence from us states. *The Economic Journal*, 117(521):F189–F217.

Banco de Portugal Microdata Research Laboratory (2025). Central Balance Sheet Harmonized Panel. dataset.

Baqaee, D. R. and Farhi, E. (2020). Productivity and misallocation in general equilibrium. *The quarterly journal of economics*, 135(1):105–163.

Bentolila, S. and Bertola, G. (1990). Firing costs and labour demand: how bad is eurosclerosis? *The Review of Economic Studies*, 57(3):381–402.

Biondi, F., Inferrera, S., Mertens, M., and Miranda, J. (2026). Declining job reallocation in europe: The role of shocks, market power, and technology. *Working paper*.

Caballero, R. J., Engel, E. M. R. A., and Haltiwanger, J. (1997). Aggregate employment dynamics: Building from microeconomic evidence. *American Economic Review*, 87(1):115–137.

Caggese, A., Guler, O., Mariathasan, M., and Mulier, K. (2022). Firing costs and productivity: Evidence from a natural experiment. *Available at SSRN*.

Cingano, F., Leonardi, M., Messina, J., and Pica, G. (2016). Employment protection legislation, capital investment and access to credit: evidence from italy. *The Economic Journal*, 126(595):1798–1822.

Cooper, R., Gong, G., and Yan, P. (2018). Costly labour adjustment: general equilibrium effects of china's employment regulations and financial reforms. *The Economic Journal*, 128(613):1879–1922.

Cooper, R., Haltiwanger, J. C., and Willis, J. (2024a). Declining responsiveness at the establishment level: Sources and productivity implications. Technical Report 32130, National Bureau of Economic Research.

Cooper, R., Horn, C.-W., and Indraccolo, L. (2024b). Covid and productivity in europe: A responsiveness perspective. *European Economic Review*, 163:104655.

De Loecker, J., Eeckhout, J., and Unger, G. (2020). The rise of market power and the macroeconomic implications. *The Quarterly Journal of Economics*, 135(2):561–644.

De Loecker, J. and Syverson, C. (2021). An industrial organization perspective on productivity. In *Handbook of industrial organization*, volume 4, pages 141–223.

De Loecker, J. and Warzynski, F. (2012). Markups and firm-level export status. *American economic review*, 102(6):2437–2471.

Decker, R. A., Haltiwanger, J., Jarmin, R. S., and Miranda, J. (2016). Declining business dynamism: What we know and the way forward. *American Economic Review*, 106(5):203–207.

Decker, R. A., Haltiwanger, J., Jarmin, R. S., and Miranda, J. (2020). Changing business dynamism and productivity: Shocks versus responsiveness. *American Economic Review*, 110(12):3952–3990.

Hamermesh, D. S. (1989). Labor demand and the structure of adjustment costs. *American Economic Review*, 79(4):674–689.

Hopenhayn, H. and Rogerson, R. (1993). Job turnover and policy evaluation: A general equilibrium analysis. *Journal of political Economy*, 101(5):915–938.

Hopenhayn, H. A. (1992). Entry, exit, and firm dynamics in long run equilibrium. *Econometrica*, pages 1127–1150.

Hsieh, C.-T. and Klenow, P. J. (2009). Misallocation and manufacturing tfp in china and india. *The Quarterly journal of economics*, 124(4):1403–1448.

Ilut, C., Kehrig, M., and Schneider, M. (2018). Slow to hire, quick to fire: Employment dynamics with asymmetric responses to news. *Journal of Political Economy*, 126(5):2011–2071.

Martins, F. (2019). *Portuguese Economic Growth: A View on Structural Features, Blockages and Reforms*, chapter 3.3. What did the reform of severance pay bring?, pages 105–112. Banco de Portugal.

OECD (2017). Labour market reforms in portugal 2011–2015, a preliminary assessment.

Santos, C. D., Costa, L. F., and Brito, P. B. (2022). Demand, supply and markup fluctuations. *The Economic Journal*, 132(644):1620–1645.

Statistics Portugal (2025). Inquérito Anual à Produção Industrial. dataset.

Varejão, J. and Portugal, P. (2007). Employment dynamics and the structure of labor adjustment costs. *Journal of Labor Economics*, 25(1):137–165.

Costs of Adjustment and Firms' Responsiveness: Evidence from a Labour Market Reform

Supplemental Appendix

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Abstract

This appendix is organised as follows. Appendix A describes the institutional details of the labour market reform. Appendix B derives the revenue function from the primitives presented in the main text, and provides evidence on the potential causes behind the estimated fall in its curvature. Appendix C proves the reported increase in employment responsiveness is consistent with alternative data generating processes. Lastly, Appendix D presents additional results stemming from the structural estimation of the model.

A Institutional setting: the labour market reform in Portugal

The Portuguese labour market ranked as the most stringent across OECD countries prior to the GFC and the ensuing Euro Area crisis. Between 2008 and 2019 that strictness would fall and be brought much closer to the OECD unweighted average. This striking evolution is seen in both panels of Figure 2 in the main text.

What can then explain this move from first to ninth? In April 2011 the Portuguese government requested financial assistance to the European Commission, the European Central Bank, and the International Monetary Fund. The memorandum of understanding signed a month later tied a bailout package of 78 billion euros to a series of public spending cuts and significant reforms in the economy. The most important ones were arguably those pertaining to the labour market.

Indeed, over the following years lawmakers introduced multiple changes to the Portuguese labour law. A comprehensive overview of all those, as well as their intended goals, can be found in OECD (2017). Yet my emphasis in the paper is placed on three very specific policies which were meant to reduce the costs borne by firms when letting go employees: *i*) severance payments were slashed, *ii*) firm-worker funds were set up in order to accommodate up to half of severance due in case of dismissal, and *iii*) a new and more flexible justification for fair dismissals was created. I go over these in detail.

Start with the severance payments cut. Initially, for every full year of tenure at the firm, the severance pay of workers on open-ended contracts (2/3 of all employment relationships) corresponded to 30 days of base wage, plus tenure-based increments. A minimum of three months' wage would have to be paid and no

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maximum would apply. All these conditions were eventually scrapped: lawmakers progressively reduced the amount of severance to only 12 days of base wage (from 30 to 20 in November 2011 and ultimately to just 12 in October 2013); the minimum was removed, and a cap of 240 times the national monthly minimum wage was introduced. Similar provisions were also set for fixed-term contracts, albeit with additional conditional clauses on the length of the terminated contract.

This was not the only change made to severance payments. In an attempt to prevent liquidity shortages caused by the costs borne at the time of dismissal, firms were required to constitute worker-specific funds. This provision was instituted in October 2013 and encompasses no risks for employers. Each month they make a mandatory contribution of 0.925% of each employees' wage to their respective fund. Should a worker leave the firm voluntarily, no severance is due and the firm is paid back all the contributions made so far. On the other hand, if the worker is entitled to severance, then the fund covers half of that amount.

Lastly, the reform also introduced a more flexible procedure to justify the fair dismissal of an employee who is revealed to be unsuitable for the job. Before August 2012 a worker could only be let go on the grounds of unsuitability after a lengthy series of steps had been painstakingly followed: first, the employer would have to prove that the worker's inability had been preceded by a change in the nature of the assigned tasks; conditional on those changes, the employer would then have to demonstrate that *i*) the lack of ability had not been remedied by the provision of additional and certified training, *ii*) the worker could not be transferred to a more suitable role within the company, and *iii*) the situation had not been generated by a lack of health and safety conditions for which the employer could be held accountable. The process has been significantly shortened. First, it can now start even when the nature of the tasks remains unchanged. And second, the employer is no longer required to offer an alternative and suitable position within the firm.

Two final remarks about the policies described here. First, they were part of a wider labour market reform which took place in Portugal. For example, the memorandum of understanding also encompassed minimum wage freezes, easier access to less generous unemployment benefits, and changes to the system of collective bargain agreements. The ultimate goal of these policies was clear: to boost job reallocation, and eventually productivity, through a less stringent labour market. The second one concerns their immediate effectiveness. All enacted provisions on severance payments were grandfathered in, *i.e.* they would only apply to new contracts set from then on. In the main text I argued this may have mitigated the reduction in employment inaction, even though it may have produced immediate incentives for job creation.

B Revenue function

B.1 Derivation

Firms produce according to $y = z \cdot (nh)^{\zeta} \cdot k^{1-\zeta}$, and face a demand price schedule given by $p(y) = (\theta/y)^{1/\nu}$. A detailed explanation of these two functions can be found in the main text. For the sake of clarity, I am omitting time- and firm-specific subscripts.

The revenue flow net of capital payments can be defined as follows:

$$R(\cdot) \equiv \max_k \left\{ p(y) \cdot y - rk \right\} = \max_k \left\{ \theta^{\frac{1}{\nu}} \cdot \left(z \cdot (nh)^{\zeta} \cdot k^{1-\zeta} \right)^{\frac{\nu-1}{\nu}} - rk \right\},$$

where r corresponds to exogenous price of capital, *i.e.* the real interest rate. For convenience of notation let $\varepsilon \equiv (\nu - 1)/\nu$. Under the assumption of fully flexible capital taking and developing the associated FOC is necessary and sufficient to yield the following demand function:

$$k = \left(\frac{\varepsilon(1-\zeta) \cdot \theta^{1-\varepsilon} \cdot \left(z(nh)^{\zeta} \right)^{\varepsilon}}{r} \right)^{\frac{1}{1-\varepsilon(1-\zeta)}}$$

Plugging this solution back in the objective function allows to retrieve a revenue function which is entirely expressed in terms of labour demand, and the stochastic evolution of technological and demand shocks:

$$\begin{aligned} R(\cdot) &= \theta^{1-\varepsilon} \left(z(nh)^{\zeta} \right)^{\varepsilon} \cdot \left(\frac{\varepsilon(1-\zeta) \cdot \theta^{1-\varepsilon} \cdot \left(z(nh)^{\zeta} \right)^{\varepsilon}}{r} \right)^{\frac{\varepsilon(1-\zeta)}{1-\varepsilon(1-\zeta)}} - r \left(\frac{\varepsilon(1-\zeta) \cdot \theta^{1-\varepsilon} \cdot \left(z(nh)^{\zeta} \right)^{\varepsilon}}{r} \right)^{\frac{1}{1-\varepsilon(1-\zeta)}} \\ &= \theta^{\frac{1-\varepsilon}{1-\varepsilon(1-\zeta)}} \left(z(nh)^{\zeta} \right)^{\frac{\varepsilon}{1-\varepsilon(1-\zeta)}} \cdot \left(\left(\frac{\varepsilon(1-\zeta)}{r} \right)^{\frac{\varepsilon(1-\zeta)}{1-\varepsilon(1-\zeta)}} - \left(\varepsilon(1-\zeta) \right)^{\frac{1}{1-\varepsilon(1-\zeta)}} \cdot \left(\frac{1}{r} \right)^{\frac{\varepsilon(1-\zeta)}{1-\varepsilon(1-\zeta)}} \right) \\ &= \underbrace{\left(1 - \varepsilon(1-\zeta) \right) \cdot \left(\frac{\varepsilon(1-\zeta) \cdot z^{\frac{1}{1-\zeta}}}{r} \right)^{\frac{\varepsilon(1-\zeta)}{1-\varepsilon(1-\zeta)}} \cdot \theta^{\frac{1-\varepsilon}{1-\varepsilon(1-\zeta)}}}_{\equiv A(z,\theta,r)} \cdot \underbrace{(nh)^{\frac{\zeta\varepsilon}{1-\varepsilon(1-\zeta)}}}_{\equiv (nh)^{\alpha}} \end{aligned} \tag{B1}$$

The reduced form of the revenue function is simply given by $R(A, n, h) \equiv A \cdot (nh)^{\alpha}$. From equation (B1) it is clear the profitability level, A, encompasses changes to productivity (z), demand (θ), and the cost of capital (r). It also includes various parameters of the production and demand functions. The curvature, on the other hand, corresponds to $\alpha \equiv \zeta\varepsilon / \left(1 - \varepsilon(1-\zeta) \right)$. As pointed out in the main text, it embeds both the labour elasticity in the production function (ζ), and the price elasticity of demand ($\varepsilon \equiv (\nu - 1)/\nu$).

B.2 Aggregate measure of markups

Let $C(y)$ be the total cost function of the firm, such that profits can be written as follows:

$$\pi(y) \equiv y \cdot p(y) - C(y) = y \cdot \left(\frac{\theta}{y} \right)^{1/\nu} - C(y)$$

A simple FOC with respect to y yields the following:

$$\left(\frac{\theta}{y}\right)^{1/\nu} \times \left(\frac{\nu-1}{\nu}\right) = \frac{\partial C(y)}{\partial y} \tag{B2}$$

All three components of condition (B2) above are known. The one on the right-hand side is the marginal cost, $mc(y) \equiv \partial C(y)/\partial y$. The product of the left-hand side comprises the price schedule faced by firms, $p(y)$, and a composite parameter I had already labelled $\varepsilon \equiv (\nu-1)/\nu$. The associated markup, μ, is the ratio between price and marginal cost, which then corresponds to the following:

$$\mu \equiv \frac{p(y)}{mc(y)} = \frac{\nu}{\nu-1} = \varepsilon^{-1} \tag{B3}$$

B.3 Disentangling the fall in α

Table 1 documents a statistically significant fall in the curvature of firms' revenue function. Although I refrain from pinpointing its underlying cause, sub-section 2.5 in the main text discusses potential economic forces behind it. I now provide tentative evidence in support of the various channels proposed. I start by discarding the hypothesis that the reduction in α could be the outcome of a changing sectoral composition in the Portuguese economy. This suggests a more fundamental mechanism is at play. I investigate two plausible ones: an increase in product market power, and broader changes in the demand structure of the economy impacting the effective elasticity of demand faced by firms.

Sectoral decomposition of the economy

An initial concern is that the reduction in α could have actually reflected an endogenous reshuffle of employment towards firms and sectors whose curvature was already smaller in the first place. I reject that hypothesis with two pieces of evidence. The first one is shown in Table B1 below and it displays the evolution of employment shares. Between 2008 and 2019 the ranking of the five largest sectors has remained unchanged: Manufacturing, Wholesale and retail trade, Construction, Administrative and supportive services, and Accommodation and food services jointly account for roughly three quarters of all employment in non-financial firms. The only noticeable change concerns the construction sector. Before the detrimental effects of the GFC kicked in, approximately 13.7% of all employees worked in construction. Eleven years later, that figure only amounted to 9.16%.

In a second exercise I estimate sectoral revenue functions. More specifically, I repeat the estimation exercise of equation (1), but this time I do it for each of the five largest sectors in the economy. The goal is to understand whether these sector-specific curvatures mimic the behaviour of the aggregate one. The results are displayed in Table B2 below. The fall is observed in all five sectors. And just as interestingly, all those drops have similar orders of magnitude. According to Table 1 in the main text, the curvature of the revenue function for the whole economy dropped 21%, from 0.56 to 0.44. For the sectoral ones that value ranges between 22% (Administrative and supportive services) and 30% (Accommodation and food services).

Table B1: Sectoral employment shares (%)

2-digit industry	**2008**	**2019**
Manufacturing	25.37	22.39
Wholesale and retail trade	21.93	20.89
Construction	13.73	9.16
Administrative and supportive services	9.69	11.23
Accommodation and food services	7.23	9.23
Transportation and warehousing	5.59	5.48
Consulting, scientific and technical services	4.33	5.79
Information and communication services	2.34	3.57
Health care and social assistance	2.30	3.56
Agriculture, forestry, hunting and fishing	1.54	2.43
Educational services	1.35	1.35
Real estate	1.06	1.39
Other services	1.02	0.99
Water, sewage and waste management	0.89	0.94
Arts, entertainment and recreation	0.62	0.97
Extractive industries	0.45	0.28
Electricity	0.35	0.33
Finance and insurance	0.16	0.02
Public administration	0.04	0.00
Production activities for self-consumption	0.00	0.00
Observations	274 485	327 680

Fall in α caused by greater product market power

Under the framework set up in Appendix B.1 the curvature of the revenue function is pinned down by two primitive parameters: the output elasticity of labour (ζ), and the price elasticity of demand ($\varepsilon \equiv (\nu - 1)/\nu$).

$$\alpha \equiv \frac{\zeta\,\varepsilon}{1 - \varepsilon(1-\zeta)} = \frac{\zeta}{\mu + \zeta - 1}, \tag{B4}$$

Table B2: Sectoral estimation of the curvature

2-digit industry	**Employment share (%)**	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Manufacturing	24.30	0.653^{***}	0.495^{***}
		(0.017)	(0.020)
Wholesale and retail trade	21.93	0.648^{***}	0.467^{***}
		(0.012)	(0.012)
Construction	10.90	0.969^{***}	0.706^{***}
		(0.018)	(0.026)
Administrative and supportive services	10.40	0.611^{***}	0.474^{***}
		(0.028)	(0.028)
Accommodation and food services	7.97	1.090^{***}	0.767^{***}
		(0.022)	(0.021)

Notes: Parameter α is estimated directly from equation (1). The corresponding standard errors are robust. The number of observations for the full period, pre-reform, and post-reform are respectively as follows: *Manufacturing* has 480 369, 204 430, and 141 962; *Wholesale and retail trade* has 1 124 121, 404 030, and 288 223; *Construction* has 444 120, 178 376, and 110 635; *Administrative and supportive services* has 135 198, 41 684, and 35 069; and *Accommodation and food services* has 386 467, 136 391, and 113 758.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

where the equality comes from the definition of the aggregate markup displayed in equation (B3). Given there is a one-to-one mapping between the price elasticity of demand and the aggregate markup, this particular setting predicts that changes in the curvature α ultimately reflect changes in ζ and/or μ.

In Table 4 I showed the estimated fall in the curvature was mostly driven by the fact that the isoelastic demand schedule faced by firms became increasingly inelastic. As consumers become more insensitive to price changes, firms can charge higher markups. In this narrative, the reduction in the curvature of the revenue function is caused by an overall increase in product markups, an evolution compatible with a trend documented by De Loecker and Eeckhout (2018) for most North American and European economies. It is also consistent with a poiny recently made by Biondi et al. (2026): markups increased in the set of European countries which experienced a decline in employment responsiveness.

Yet this finding relies heavily on the framework of the economy: firms are assumed to produce according to a constant returns to scale technology, and face an isoelastic demand schedule. But it also hinges on the ability to recover the output elasticity of labour as the an average of the labour share across producers and time (De Loecker and Syverson (2021)). The result holds when I repeat the averaging procedure with a caveat: I take a *weighted* mean of firms' labour share — see Table B3. The existence of a production function with constant returns to scale can also be shown to be an uncontroversial premise. In their attempt to study the cyclicality of markups in the Portuguese economy, Santos et al. (2022) estimate (industry-wide) Cobb-Douglas and translog production functions using a large firm-product panel of firms between 2004 and 2014. With Cobb-Douglas, the sum of coefficients in the production functions ranges between 0.68 and 1.10; the null hypothesis that sum equals 1 is not rejected at 10% in half of the selected industries. This figure is even greater when a translog specification is assumed. In that case, the authors do not reject the existence of constant returns to scale in fourteen out of the eighteen industries.

Table B3: Disentangled changes in the estimated curvature of the revenue function (weighted mean)

	Pre-reform: 2006-2013	**Post-reform: 2014-2019**
Curvature, α	0.563	0.442
Output elasticity of labour, ζ	0.649	0.741
Implied elasticity of demand, $\nu \equiv \lvert(1-\varepsilon)^{-1}\rvert$	2.988	2.071
Implied markup, $\mu \equiv \varepsilon^{-1}$	1.503	1.934
Counterfactual α: constant labour elasticity		-7.339%
Counterfactual α: constant markup		$+34.65\%$

Notes: The curvature of the revenue function corresponds to $\alpha \equiv \frac{\zeta\varepsilon}{1-\varepsilon(1-\zeta)}$. The labour share of a firm is computed as the ratio between the wage bill and its revenues. The aggregate labour elasticity ζ is obtained by taking the employment-weighted mean of (1% winsorized) firm-level values and then the time-series average across the corresponding time window. The number of observations in each time window is the same as reported in Table 1.

However, in order to produce more substantive evidence on this channel, I follow the seminal work of De Loecker and Warzynski (2012) and identify how the underlying distribution of firm-level markups evolved over time. At its core, this method posits that markups can be retrieved as the wedge between an input's output elasticity and its revenue share. But not any input. Consider a firm which engages in cost minimisation and uses an input m which satisfies the following assumptions:

Assumption I: There are no adjustment costs for input m.

Assumption II: Input m is chosen statically.

Assumption III: Input m is not subject to monopsony forces.

Assumption IV: Input m is used for the production of output only.

If assumptions I-IV do hold, then the firm-level markup can be computed as follows:

$$\mu_{i,x,t} = \frac{\beta^m_{x,t}}{s^m_{i,x,t}}, \tag{B5}$$

where $\beta^m_{x,t}$ is the output elasticity of input m in sector x period t, and $s^m_{i,x,t}$ is the revenue share of firm i in sector x in period t. For what follows, I take intermediate materials to take on the role of input m. Two key questions are immediately raised concerning identification. The first one is whether those intermediates do lack adjustment costs and monopsony power. Yeh et al. (2022) provide such discussion in the context of the US. The second one relates to the estimation of $\beta^m_{x,t}$. Unlike $s^m_{i,x,t}$, this is an object which cannot be read straight off the data. To do so I rely on Ackerberg et al. (2015) to estimate revenue functions at the 2-digit sectoral level, and which take labour, capital, and intermediate inputs as arguments. Although a more detailed explanation can be found in Appendix C, this procedure comes at the expenses of the following assumption:

Assumption V: The demand function for input m is strictly monotone in unobserved productivity.

Given the estimated values of $\beta^m_{x,t}$ and the observed values of $s^m_{i,x,t}$ one can compute the product markup and follow its distribution over time. Some descriptive statistics are described in Table B4 below. The distribution of firm-level markups is indeed consistent with an increase in product market power exerted by firms: over time, both the mean and median have been shifting to the right.

Table B4: Markup statistics

	Mean	Std. deviation	p25	p50	p75	Observations
2006-2008	2.52	3.77	0.87	1.19	2.02	612 647
2006-2013	2.58	3.89	0.87	1.21	2.05	1 627 782
2014-2019	2.73	4.17	0.98	1.28	2.04	1 255 966

Notes: The firm-level markup is computed as $\mu_{i,t} = \beta^m_{i,t}/s^m_{i,t}$, where $\beta^m_{i,t}$ is the revenue elasticity of materials, and $s^m_{i,t}$ its revenue share. Both the revenue share of materials and the markup have been winsorized at the 5% tails.

A final remark on this exercise is in order. The markup estimator defined in equation (B5) requires that a measure of the *output* elasticity of input m be used. Yet physical output is not observed in the dataset. Instead of a production function, I am using Ackerberg et al. (2015) to actually identify a *revenue* elasticity of input m. Bond et al. (2021) claim that estimated markups built on deflated revenues, as is the case here, must inevitably equal unity. If they do not, as is also the case, then one of assumptions I-IV is not held. Their critique implies that my markup estimates above are biased. The point is taken but does not alter the main message I want to convey. This procedure is not meant to accurately identify the values of markups in the Portuguese economy but simply to document a trend: market power exerted by firms has risen in the second half of the sample. According to De Ridder et al. (2026), revenue-based estimates can still capture that dynamics.

Fall in α caused by broader changes in demand structure

Attributing the fall in α to rising product market power exerted by firms rests on a large number of assumptions. Although Santos et al. (2022) cannot rule out the existence of constant returns to scale technology in the Portuguese economy, their estimates do seem to question the premise of an iselastic demand schedule. The existence of varying markups reported in Table B4 before is also inconsistent with that assumption.

This implies the true reason for the decline in the estimated values of α may actually lie elsewhere. In this sub-section I look at broader changes in the demand structure of the economy which could plausibly account for the observed change in the curvature. In the original framework I assumed firms faced a unique isoelastic demand schedule. Suppose now the reported revenue is in fact an aggregation of sales in two different markets with potentially different elasticities.

$$R(\cdot) = \max_{y>0, y^*\geq 0} \left\{ y \cdot \left(\frac{\theta}{y}\right)^{1/\nu} + y^* \cdot \left(\frac{\theta^*}{y^*}\right)^{1/\nu^*} \right\}, \text{ subject to } \; y + y^* \leq z \cdot n^{\zeta} \cdot k^{1-\zeta} \tag{B6}$$

Firms must assign production to each market subject to a capacity constraint. Amongst other things, the problem in (B6) is suitable to study exporting decisions of heterogeneous firms. In this case, they are assumed to have access to both *domestic* and *foreign* demand shifters — θ and θ^*, respectively. And since the elasticities may be different, $\nu \neq \nu^*$, the revenue function can no longer be simplified to a convenient expression as that of the main text. However, if this is indeed the true data generating process, the estimates of α displayed in Table 1 actually mirror a greater number of objects: the elasticities of demand in both the domestic and foreign markets, beyond the output elasticity of labour.

It is reasonable to expect that exposure to foreign demand swings has become more relevant. Table B5 below, for example, shows that while the employment shares of the five largest sectors has remained largely unchanged, their reliance on revenues from abroad has grown. A similar conclusion can be drawn from more aggregate figures: in 2006, only 13% of firms were exporters and foreign sales amounted to 11% of all revenues; by 2019, those figures had already risen to 19% and 16%, respectively. Regardless of how aggregation is effectively carried out, this rising openness of the Portuguese economy is bound to have an impact on the structure of firms' revenue function. The observed reduction in α may in fact be the outcome of this changing composition: as Portuguese firms generate a larger share of their revenues abroad, where demand elasticities may differ, a falling curvature may actually reflect deeper changes in this more complex demand system. Notice this narrative can go beyond the initial channel of greater markups. With more flexible employment adjustments, firms can more easily upscale and be able to pay the sunk cost of entering export markets (see Cooper et al. (2026)).

Alternatively, one can also conceive that the reduced frictions in the labour market facilitate firms' decision to add or drop products to their portfolio. Basically, suppose instead that (B6) captures how firms'

Table B5: Sectoral shares (%) of employment and foreign sales

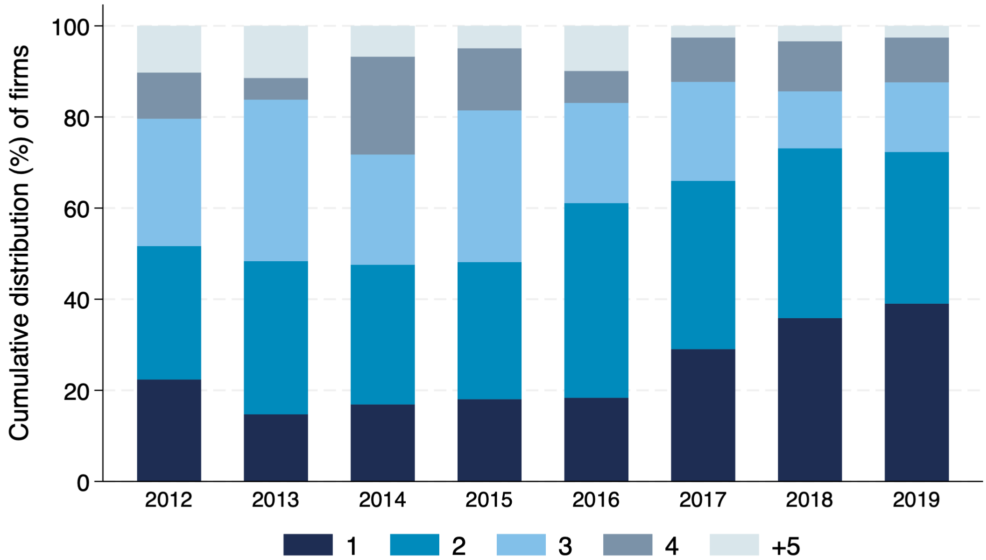
2-digit industry	**2008**		**2019**	
	Employment	Foreign sales	Employment	Foreign sales
Manufacturing	25.37	30.89	22.39	37.33
Wholesale and retail trade	21.93	6.47	20.89	8.44
Construction	13.73	6.74	9.16	9.11
Administrative and supportive services	9.69	8.68	11.23	15.39
Accommodation and food services	7.23	2.85	9.23	3.76

Notes: The employment share corresponds to the total number of people employed in each 2-digit industry divided by the aggregate employment of the economy in the corresponding year. The share of foreign sales is the amount of foreign revenue produced by each 2-digit industry divided by its overall revenue in that year. The number of observations for 2008 and 2019 is respectively as follows: *Manufacturing* has 35 831 and 34 747; *Wholesale and retail trade* has 80 964 and 81 720; *Construction* has 35 395 and 34 095; *Administrative and supportive services* has 8870 and 12 249; and *Accommodation and food services* has 25 123 and 33 476.

revenue is generated in various *product* markets. This is also a considerable margin of adjustment. According to Figure B1 below, more than half of all Portuguese manufacturing firms produce more than one product. And just as importantly, there has been substantial dynamism in that choice: while in 2012 only 22% of manufacturing businesses were single-product, by 2019 that share had almost doubled to 39%. Such an evolution can again be rationalised as heterogeneous firms deciding to enter multiple markets where elasticities may be different. In this particular case, the fall in α can be viewed as encompassing changes in the effective elasticity of demand faced by firms, and which stems from operations in different product markets.

In conclusion, the estimated change in the curvature of the revenue function need not reflect a shift in the distribution of markups. In fact, broader changes in the demand struture of the economy may be driving that observed fall. Here I just hinted at two possible forces: the increasing openness of Portuguese firms and their rising reliance on one-product sales are not necessarily markup developments, but can justify the reduction in α through changes in the elasticities of substitution across the various markets in which firms participate.

Figure B1: Distribution of the number of products produced by Portuguese manufacturing firms



Notes: The underlying data come the IAPI databse (see Statistics Portugal (2025)), a yearly survey on Portuguese manufacturing firms. The sample contains information on revenues and quantities produced and sold at a detailed 9-digit product level. The number of unique firms from 2012 on evolves as follows: 44 627 in 2012; 42 522 in 2013; 41 604 in 2014; 41 068 in 2015; 40 731 in 2016; 43 248 in 2017; 43 431 in 2018; and 44 624 in 2019. Data before 2012 are available but were excluded due to a reduced number of observations.

C Additional empirical facts

C.1 Dataset

The dataset is the Central Balance Sheet Harmonized Panel (CBHP), maintained by the Banco de Portugal Microdata Research Laboratory (2025). It contains yearly data since 2006 on the numerous elements of non-financial firms' balance sheets and income statements. The data are collected through *Informação Empresarial Simplificada*, a declaration firms are required by law to submit to the national tax authority until July 15$^{\text{th}}$ every year. Given its mandatory nature, the coverage of the data is impressive: more than 600 000 unique firms across the entire sample. Table C1 shows this and other relevant statistics.

The table reiterates some of the points which had already been made. First, the distribution of labour productivity, as measured by the natural logarithm of revenue per worker, remained largely unchanged. Both the mean and the standard deviation do not exhibit any noticeable swings. Second, the same can be said about the employment-size distribution. Notwithstanding the changes in the labour law, the average sized Portuguese firm employs 9 workers, while the one at the median has 6.

Table C1: Summary of statistics

		Full sample: 2006-2019	Pre-reform: 2006-2013	Post-reform: 2014-2019
Observations		3 983 356	2 182 179	1 801 177
Nr. unique firms		616 441	445 447	435 628
Employees	Mean	9.41	9.56	9.22
	p25	1.00	2.00	1.00
	p50	3.00	3.00	3.00
	p75	6.00	6.00	6.00
Log revenue pw	Mean	10.60	10.63	10.57
	Std. deviation	1.15	1.14	1.16
Exit rate (%)	Mean	7.42	8.16	6.44

C.2 Robustness checks on the employment policy function

The excessive inaction

The pervasiveness of employment inaction had already been documented in Varejão and Portugal (2007). An additional and very important feature of the Portuguese labour market concerns the widespread usage of collective bargain agreements. According to Card and Cardoso (2022), 87% of all private sector full-time workers were covered by one of those — see also Addison et al. (2017). These agreements can regulate various aspects of the employment relationship, chiefly amongst them the minimum wage level for the different job titles.

This constraint on wage adjustments naturally compounds the effect exerted by the non-convexities studied in the main text. As such, the exceedingly large values of inaction reported in Figure 3 may actually indicate that more than one lag should be included in the employment policy function. I now consider the alternative specification:

$$\log(n_{i,t}) = \tilde{\lambda}_1 \log(n_{i,t-1}) + \tilde{\lambda}_2 \log(n_{i,t-2}) + \tilde{\lambda}_3 \log(\xi_{i,t}) + \epsilon_{i,t} \tag{C1}$$

Responsiveness is now measured by $\partial \log(n_{i,t})/\partial \log(\xi_{i,t}) = \tilde{\lambda}_3$. The estimates are shown in Table C2 below.

The main conclusions remain largely unchanged. Most importantly, there is an unequivocal increase in the employment responsiveness of firms. In the main formulation explored in Table 2 I showed that responsiveness nearly doubled after the reform. When accounting for additional lags of employment, that

Table C2: Log linear approximation of employment policy function (two lags)

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Serial correlation order 1, $\tilde{\lambda}_1$	0.913^{***}	0.911^{***}	0.913^{***}
	(0.012)	(0.012)	(0.012)
Serial correlation order 2, $\tilde{\lambda}_2$	0.056^{***}	0.053^{***}	0.062^{***}
	(0.011)	(0.011)	(0.011)
Responsiveness, $\tilde{\lambda}_3$	0.017^{**}	0.008	0.034^{***}
	(0.008)	(0.007)	(0.008)
Industry & Year FE	Yes	Yes	Yes
R^2	0.966	0.966	0.967
Observations	2 761 799	1 325 349	985 817

Notes: Standard errors are clustered at the industry level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

jump is actually fourfold.

Non-flexible capital

A contentious assumption concerning the identification of the unobserved profitability innovations in (1) is the one about the absence of frictions in capital adjustments. Since that identification scheme ultimately impacts how the responsiveness coefficient λ_2 is estimated, I now turn to this issue.

In Appendix B.1 I showed how the existence of fully flexible capital allows to express the revenue function solely in terms of labour demand. Let me now relax this assumption. Basically, say capital is not a static choice, *i.e.* it cannot be freely adjusted. This is consistent with the existence of time to build and adjustment costs. I remain agnostic about the potential causes; the main point is that the revenue function can no longer be simplified to the same expression. As a consequence, the profitability innovations must be identified with an alternative identification scheme which can account for the dynamic nature embedded in capital choices.

I do so via the procedure in Ackerberg et al. (2015). Suppose the revenue function of firm i in period t reads as follows:

$$\log(R_{i,t}) = \log(\tilde{A}_{i,t}) + \alpha_n \log(n_{i,t}) + \alpha_k \log(k_{i,t}) + \alpha_m \log(m_{i,t}) + \epsilon_{i,t} \tag{C2}$$

I addition to labour ($n_{i,t}$) and capital ($k_{i,t}$), I also consider intermediate inputs ($m_{i,t}$) as an argument of the revenue function. The profitability level $\log(\tilde{A}_{i,t})$ is assumed to follow a similar AR(1) process to the one described in the main text: the serial correlation is $|\tilde{\rho}| < 1$ and the underlying i.i.d. gaussian innovations $\log(\tilde{\xi}_{i,t})$. The usage of the tilde in these three objects is meant to reinforce the idea that this profitability process is identified with a different method of that of equation (1). There are three additional assumptions which must be emphasised. First, the choice of hours was dropped for convenience. Second, identifying the various elasticities in (C2) requires that **Assumption V** defined in Appendix B be held: the demand function for input intermediate inputs must be strictly monotone in the unobserved profitability level. And third, the estimation is ultimately carried out through GMM-IV. The vector of instruments is $[\log(k_{i,t})\ \log(n_{i,t-1})\ \log(m_{i,t-1})\ \hat{\Phi}_{t-1}(k_{i,t-1}, n_{i,t-1}, m_{i,t-1})]'$, where $\hat{\Phi}_{t-1}(\cdot)$ corresponds to the lag of the fitted values of revenue on a polynomial of order 3 in capital, employment, and intermediate inputs. A full set of year dummies also controls for the existence of aggregate shocks.

This estimation procedure allows to obtain estimates of all elasticities in (C2), as well as a non-parametric estimate of $\tilde{\rho}$. Ultimately, all these objects are sufficient to back out the series of unobserved profitability innovations without the need to appeal to full flexibility of capital demand. A slightly modified version of the employment policy function can then be set up as follows:

$$\log(n_{i,t}) = \hat{\lambda}_1 \log(n_{i,t-1}) + \hat{\lambda}_2 \log(k_{i,t-1}) + \hat{\lambda}_3 \log(\tilde{\xi}_{i,t}) + \epsilon_{i,t} \tag{C3}$$

Table C3 below displays its estimates. The results are again aligned with those reported in Table 2. First, the inclusion of the lagged stock of capital[7] does not significantly improve an already large value of R^2. And lastly, despite the larger confidence bands, this alternative specification still predicts a sizeable increase in employment responsiveness.

Table C3: Log linear approximation of modified employment policy function

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Lag employment, $\hat{\lambda}_1$	0.942***	0.935***	0.951***
	(0.005)	(0.006)	(0.004)
Lag capital, $\hat{\lambda}_2$	0.013***	0.015***	0.011***
	(0.002)	(0.002)	(0.002)
Responsiveness, $\hat{\lambda}_3$	0.050	0.074	0.126***
	(0.061)	(0.053)	(0.031)
Industry & Year FE	Yes	Yes	Yes
R^2	0.971	0.969	0.973
Observations	2 082 050	1 148 154	781 895

Notes: Standard errors are clustered at the industry level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

[7] The results would have been similar if I had instead considered that the *contemporaneous* stock of capital, and not its *lagged* one.

Non-linearities

As an additional robustness exercise I also include non-linear components to the employment policy function. The general specification reads as follows:

$$\log(n_{i,t}) = \check{\lambda}_1 \log(n_{i,t-1}) + \check{\lambda}_2 \log(\xi_{i,t}) + \check{\lambda}_3 \cdot \pi_{i,t} + \epsilon_{i,t}, \tag{C4}$$

The focus lies particularly on the third regressor, $\pi_{i,t}$, which is considered to take two different forms:

$$\pi_{i,t} = \begin{cases} \log(\xi^2_{i,t}) & \textbf{specification I} \\ \mathbb{I} \cdot \big(\log(\xi_{i,t}) < 0\big) & \textbf{specification II} \end{cases}$$

For **specification I** I introduce the quadratic component of the profitability innovation. The results are shown in Table C4 below. The policy function is concave in the profitability innovations, $\check{\lambda}_3 < 0$, both in the full sample and in the pre-reform period. Yet that concavity vanishes in the post-reform period. This increasing linearity, which had already been found at the intensive margin of employment adjustment, is also consistent with growing responsiveness.

Table C4: Log linear approximation of employment policy function: **specification I**

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Serial correlation, $\check{\lambda}_1$	0.984^{***}	0.975^{***}	0.995^{***}
	(0.003)	(0.004)	(0.002)
Responsiveness, $\check{\lambda}_2$	0.030^{***}	0.023^{***}	0.048^{***}
	(0.008)	(0.008)	(0.010)
Quadratic, $\check{\lambda}_3$	-0.003^{**}	-0.006^{***}	0.002
	(0.001)	(0.001)	(0.001)
Industry & Year FE	Yes	Yes	Yes
R^2	0.958	0.958	0.959
Observations	3 308 659	1 709 959	1 350 617

Notes: Standard errors are clustered at the industry level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

For **Specification II**, on the other hand, I attempt to capture potentially different slopes for positive and negative profitability innovations. This replicates one of the exercises reported in Ilut et al. (2018) and aims to study the existence of asymmetries in the employment policy function. Do firms react more to bad shocks than to positive ones? According to Table C5 below they do. This is wholly encapsulated in the persistently positive values of $\check{\lambda}_3$. Notice, however, that asymmetry is attenuated in the post-reform period.

Table C5: Log linear approximation of employment policy function: **specification II**

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Serial correlation, $\check{\lambda}_1$	0.955^{***}	0.951^{***}	0.961^{***}
	(0.005)	(0.006)	(0.005)
Responsiveness, $\check{\lambda}_2$	0.048^{***}	0.045^{***}	0.061^{***}
	(0.007)	(0.007)	(0.010)
Bad news, $\check{\lambda}_3$	0.028^{***}	0.035^{***}	0.021^{**}
	(0.009)	(0.011)	(0.009)
Industry & Year FE	Yes	Yes	Yes
R^2	0.959	0.958	0.959
Observations	3 308 659	1 709 959	1 350 617

Notes: Standard errors are clustered at the industry level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

The responsiveness formulation of Decker et al. (2020)

A meaningful contribution of this paper is to show that employment responsiveness of Portuguese firms rose after a comprehensive labour market reform slashed costs of adjustment. This contrasts with recent developments in the US economy reported by Decker et al. (2020): there, responsiveness has actually been declining. For an easier comparison of results, I now implement their specification. Consider the following formulation for employment growth at the firm-year level:

$$\varkappa_{i,t} = \beta_0 + \beta_1 \log(n_{i,t}) + \beta_2 \log(\xi_{i,t}) + \mathcal{T}\big(\log(\xi_{i,t}), t\big) + \epsilon_{i,t} \tag{C5}$$

All variables retain the same meaning as in the main text. Instead of splitting the sample into various sub-periods, the evolution of responsiveness is assessed through the time operator $\mathcal{T}(\cdot)$, which is defined as follows:

$$\mathcal{T}\big(\log(\xi_{i,t}), t\big) \equiv \begin{cases} \Delta \times \log(\xi_{i,t})\, \text{Trend}_t \\ \phi \times \log(\xi_{i,t})\, \mathbf{1}_{\{t \geq 2014\}} \end{cases} \tag{C6}$$

The first element assesses the evolution through the interaction between the exogenous profitability innovation and a simple linear trend. If the associated coefficient Δ is estimated to be positive, this is interpreted as proof of an increasing responsiveness throughout the sample years. The second element, on the other hand, focuses more on the change prompted by the labour market reform. In that case, the profitability innovation is interacted with an indicator variable which equals the unity from 2014 on, and 0 otherwise. The paper argues that the reform raised the employment responsiveness of firms. This implies that $\phi > 0$. Table

C6 shows the results, not just for the rowth but also the level of employment.

Table C6: Evolution of employment responsiveness over the period 2006-2019

	Employment growth, $\varkappa_{i,t}$		**Employment level**, $\log(n_{i,t})$	
Lag employment, β_1	-0.043^{***}	-0.043^{***}	0.951^{***}	0.951^{***}
	(0.005)	(0.005)	(0.005)	(0.005)
Responsiveness, β_2	0.013	0.022^{***}	0.016^{*}	0.027^{***}
	(0.007)	(0.007)	(0.008)	(0.008)
Innovation $\times$ linear trend, Δ	0.002^{***}		0.003^{***}	
	(0.000)		(0.001)	
Innovation $\times$ post-reform dummy, ϕ		0.024^{***}		0.026^{***}
		(0.004)		(0.004)
Industry & Year FE	Yes	Yes	Yes	Yes
R^2	0.030	0.030	0.904	0.904
Observations	3 060 576	3 060 576	3 060 576	3 060 576

Notes: Standard errors are clustered at the industry level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

The results are consistent with an increase in employment responsiveness: both coefficients, Δ and ϕ, are positive and statistically significant. The fact that their counterparts in Decker et al. (2020) are negative allows them to draw the conclusion that responsiveness in the US has been declining. I report the opposite for Portugal. In particular since a labour market reform was enacted, both the level and growth of labour demand became more responsive to changes in exogenous profitability circumstances.

C.3 Responsiveness regressions

Equations (3) and (4) set up the extensive and intensive margin decisions of employment adjustment. I estimated both of them and plotted their projected values against a very fine grid of log profitability innovations — see Figure 4. I now explicitly report the values of the estimated coefficients. The results for the extensive margin are shown in Table C7, while Table C8 displays those of the intensive margin. To explicitly assess the existence of potential asymmetries, in each of them I evaluate the firms' response to a ± 1 standard deviation profitability innovation. Basically, I evaluate how the probability of employment adjustment (Table C7) and the actual adjustment (Table C8) change following both a positive and a negative one-standard deviation shock to profitability. The row labeled "*Different responses?*" investigates whether those responses to positive and negative shocks are statistically different from each other.

Table C7: Extensive margin - probability of employment adjustment

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Linear, δ_1	0.015^{***}	0.014^{***}	0.021^{***}
	(0.003)	(0.003)	(0.003)
Quadratic, δ_2	0.012^{***}	0.012^{***}	0.012^{***}
	(0.003)	(0.003)	(0.003)
Lag employment, δ_3	0.176^{***}	0.175^{***}	0.178^{***}
	(0.009)	(0.010)	(0.009)
Positive response	1.143 p.p.	0.907 p.p.	1.903 p.p.
Negative response	-1.074 p.p.	-1.342 p.p.	-0.320 p.p.
Different responses?	Yes***	Yes***	Yes***
Industry & Year FE	Yes	Yes	Yes
Observations	3 308 659	1 709 959	1 350 617
R^2	0.168	0.165	0.172

Notes: Standard errors are clustered at the industry level. *Positive response* and *Negative response* evaluate the percentage point change in probability of adjustment following a positive and negative one-standard deviation shock to profitability, respectively. The row entitled *Different responses?* indicates if those responses are statistically different at various levels of significance.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Start with the extensive margin decision. The convexity of the employment adjustment hazard is justified by the constantly positive values of δ_2. Together with the estimates of δ, the joint evolution of these two parameters is indicative of a rising, yet asymmetric, responsiveness to profitability innovations. To better understand that asymmetry, consider a firm which is hit by a positive one-standard deviation shock. In the pre-reform period this would raise the probability of adjustment by approximately 0.91 percentage points; post-reform, that likelihood would increase by 1.9 percentage points. This stands in clear contrast with the probability of adjustment in the presence of a similar shock but of the opposite sign: after 2013, firms are less likely to adjust to a negative shock to their profitability. As mentioned in the main text, these results can be interpreted to a great extent in light of the duality introduced by the reform.

I now turn to the intensive margin. The concavity of the hiring rule is mirrored in the negative values of θ_2. Much like the quadratic component in the formulation of the extensive margin, its coefficient also remains fairly stable throughout the two periods. In fact, the increasing linearity of the hiring rule is mostly driven by the coefficient θ_1, which more than doubles in the post-reform period. The rising responsiveness can be attested by the size of the employment adjustments when firms are hit by one-standard deviation shocks.

Table C8: Intensive margin - employment growth of adjusters

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Intercept, θ_0	0.250^{***}	0.268^{***}	0.235^{***}
	(0.024)	(0.026)	(0.012)
Linear, θ_1	0.065^{***}	0.048^{***}	0.104^{***}
	(0.013)	(0.012)	(0.015)
Quadratic, θ_2	-0.020^{***}	-0.020^{***}	-0.017^{***}
	(0.001)	(0.001)	(0.001)
Lag employment, θ_3	-0.104^{***}	-0.097^{***}	-0.111^{***}
	(0.011)	(0.011)	(0.011)
Positive response	3.249 p.p.	2.278 p.p.	5.592 p.p.
Negative response	-4.710 p.p.	-3.844 p.p.	-6.725 p.p.
Different responses?	Yes***	Yes***	Yes***
Industry & Year FE	Yes	Yes	Yes
Observations	1 380 478	744 067	538 526
R^2	0.075	0.065	0.080

Notes: Standard errors are clustered at the industry level. *Positive response* and *Negative response* evaluate the percentage point change in employment growth following a positive and negative one-standard deviation shock to profitability, respectively. The row entitled *Different responses?* indicates if those responses are statistically different at various levels of significance.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Whether their profitability improves or worsens, in absolute value their response is always greater in the post-reform period: a positive shock can now boost employment growth by 5.6 percentage points (only 2.3 in the pre-reform period), whereas a negative one reduces growth by 6.7 percentage points (only 3.8 before).

The asymmetry is not as striking as that associated with the extensive margin decision. To formally address this issue, I run the following specification:

$$\varkappa_{i,t\{\mathbb{1}_{i,t}=1\}} = \tilde{\theta}_0 + \tilde{\theta}_1 \log(\xi_{i,t}) + \tilde{\theta}_2 \cdot \Big(\mathbb{I} \cdot \big(\log(\xi_{i,t}) < 0\big)\Big) + \tilde{\theta}_3 \log(n_{i,t-1}) + \epsilon_{i,t} \tag{C7}$$

This is one of hiring rules specified in Ilut et al. (2018). If the parameter $\tilde{\theta}_2$ is estimated to be statistically insignificant, then this corroborates the absence of an asymmetry in employment growth amongst adjusting firms. The results are displayed in Table C9 below. The asymmetry does exit and was particularly sizeable in the pre-reform period.

Table C9: Intensive margin of adjustment - employment growth of adjusters conditional on sign

	Full sample: 2006-2019	**Pre-reform: 2006-2013**	**Post-reform: 2014-2019**
Intercept, $\tilde{\theta}_0$	0.206^{***}	0.218^{***}	0.201^{***}
	(0.026)	(0.029)	(0.015)
Linear responsiveness, $\tilde{\theta}_1$	0.093^{***}	0.081^{***}	0.123^{***}
	(0.012)	(0.012)	(0.014)
Bad news, $\tilde{\theta}_2$	0.056^{**}	0.068^{***}	0.039^{**}
	(0.021)	(0.023)	(0.017)
Lag employment, $\tilde{\theta}_3$	-0.098^{***}	-0.089^{***}	-0.106^{***}
	(0.010)	(0.010)	(0.011)
Positive response	5.593 p.p.	4.892 p.p.	7.329 p.p.
Negative response	−8.982 p.p.	−8.981 p.p.	−9.631 p.p.
Different responses?	Yes**	Yes***	Yes**
Industry & Year FE	Yes	Yes	Yes
Observations	1 380 478	744 067	538 526
R^2	0.073	0.064	0.078

Notes: Standard errors are clustered at the industry level. *Positive response* and *Negative response* evaluate the percentage point change in employment growth following a positive and negative one-standard deviation shock to profitability, respectively. The row entitled *Different responses?* indicates if those responses are statistically different at various levels of significance.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

D Additional quantitative results

D.1 Alternative parameterisation

In the baseline estimation procedure described in Section 4 I used the conforming identity matrix to weight the distances between data and simulated moments. I now report something different. Following Adda and Cooper (2003) the optimal weights are provided by the elements of the inverted variance-covariance matrix of the simulated moments. This requires that a two-stage SMM procedure be followed.

Stage 1: I solve the problem in (10) using the conforming identity matrix as weights. Let the resulting vectors of parameter estimates be labelled $\hat{\Theta}_1$.

Stage 2: I simulate 1000 economies parameterised by $\hat{\Theta}_1$. For each of those simulated economies I record the vector of targeted moments. Eventually, I obtain the variance-covariance matrix of all those

1000 vectors. I invert that matrix and use it as a weight while solving (10) a second time. The vector of estimated parameters is then $\hat{\Theta}_2$.

Although the identification scheme which underlies this two-stage procedure is the same as that employed in the main text, the usage of the optimal weighting matrix allows to downweight moments which are generated less precisely in the simulation. Naturally, this has an impact on the vector of estimated parameters. For example, in Figure 5 I showed that changes in the average exit rate of the economy exert a sizeable influence on the cost of operation, c. Suppose that exit would exhibit a significant variance across the 1000 simulations. The optimal weighting matrix would attach a smaller weight to deviations between the simulated exit rate and its data counterpart, and the estimation of parameter c would inevitably be affected.

Table D1 below show that, albeit different, the vector of estimated parameters remains largely unaltered relative to the baseline case. Importantly, this alternative parameterisation is still consistent with a significant reduction in firing costs. Prior to the reform, sacking one single employee would amount to the entire average revenue in the economy. This again entails both the fixed and linear component of the firing cost. In the period after the reform, that same cost would be equivalent to 64% of the average revenue.

Table D1: Parameter estimates in a 2-stage SMM procedure

	Labour adjustment costs				Wage function		Fixed cost
	γ_f	Γ_f	γ_h	Γ_h	η	w_0	c
2006-2013	0.27	0.73	0.22	0.13	4.19	1.36	0.04
2014-2019	0.20	0.44	0.29	0.10	4.16	0.73	0.06

Notes: All labour adjustment costs parameters and the fixed cost of operation are expressed as a share of the average revenue of the simulated economy. The point estimates are obtained via a two-stage SMM procedure. The optimal weighting matrix is employed after the economy is simulated 1000 times. All simulations are run at a quarterly frequency and then aggregated to match the annual moments of the data. The quarterly discount factor is set to $\beta = 0.99$, and the hourly rate w_1 is calibrated such that the steady-state value of hours per worker replicates the observed yearly average.

Lastly, Table D2 reports the fitness of the two-stage SMM estimation. Unlike the baseline case, the measure of distance now takes exceedingly larger values. This is inevitable, given that $F(\cdot)$ is now a *weighted* sum of distances, and those weights can indeed take very large values. This notwithstanding, the estimated model is still capable of getting the vector of selected moments just right. The concavity of the hiring rule and the exit responsiveness, for example, are particularly well targeted, just like the average employment in the economy.

Table D2: SMM estimation fit — targeted moments in a 2-stage SMM procedure

		Responsiveness				Distributions		Exit	Distance
		ϱ_{An}	λ_1	θ_2	ψ_2	$\sqrt{\Sigma_{\bar{r}}}$	$\overline{n}$	$\overline{x}$	$F(\hat{\Theta}_2)$
2006-2013	*Data*	0.64	0.95	-0.02	-0.08	1.14	1.24	8.16	
	Model	0.59	0.81	-0.02	-0.08	1.80	0.98	6.31	13736.03
2014-2019	*Data*	0.81	0.96	-0.02	-0.07	1.16	1.14	6.44	
	Model	0.57	0.85	-0.01	-0.07	1.59	1.21	5.44	3153.2

Notes: Targeted moments are the covariance between the log level of TFPR and log employment ϱ_{An}, the serial correlation of log employment λ_1 in (2), the concavity of the hiring rule θ_2 in (3), the exit responsiveness coefficient ψ_2 in (5), the standard deviation of log labour productivity $\sqrt{\Sigma_{\bar{r}}}$, the mean of log employment $\overline{n}$, and the mean of the exit rate $\overline{x}$. The measure of distance between data and simulated moments defined in (10) is $F(\hat{\Theta}_2)$.

D.2 Untargeted moments

In this sub-section I report some additional results produced by the estimated model. More specifically, I focus on three moments which were left untargeted in the estimation procedure: *i)* the cross-sectional mean of labour productivity, $\mathbb{E}(\bar{r})$, *ii)* the time-series average of business dynamism as defined in Figure 1, and *iii)* the average-sized firm's unconditional probability of exiting the economy, $\Pr_A\left(V^o(\overline{n}) > V^i(A,\overline{n})\right)$. Notice the latter object does not have an observable counterpart in the data, as it is only defined in the model. For the firm with the average number of employees I compute the exit threshold level of profitability — call it $\check{A}(\overline{n})$. Since that particular firm exits whenever $A \leq \check{A}(\overline{n})$, the third moment corresponds to the ergodic unconditional probability of that happening. The values are displayed in Table D3 below.

Given these are untargeted moments, the fit between data and model is far from perfect. Yet there are still some interesting conclusions to be drawn. First, the estimated model can replicate the observed flatness of the average labour productivity. The counterfactual exercises also deliver an expected result: that cross-sectional mean would rise with an increasingly flexible labour market, but it would have fallen if adjustment costs had not been cut. Second, although the model does not include all the necessary ingredients to replicate the downward trend of business dynamism, it seems to suggest the stringency of the labour market is an important one. Changes in adjustment costs produce very sizeable swings in the predicted value of business dynamism. Finally, there are no noticeable differences in the exit threshold. According to the ergodic distribution of profitability, the firm with the average number of workers has a 5% unconditional probability of exiting each period. This figure is broadly in line with the observed exit rate of the economy: 8.16% before the reform, 6.44% afterwards.

Table D3: Counterfactual exercises on untargeted moments

		Untargeted moments		
		$\mathbb{E}(\bar{r})$	Business Dyn.	$\text{Pr}_A\left(V^o(\overline{n}) > V^i(A,\overline{n})\right)$
2006-2013	*Data*	10.63	20.43	—
	Model	2.03	9.95	0.05
2014-2019	*Data*	10.57	18.34	—
	Model	1.79	10.82	0.05
	Counterfactual, $C(n_{-1}, n)$	1.24	3.20	0.01
	Counterfactual, *free adjustments*	2.19	123.99	0.01
	Counterfactual, α	0.49	5.09	0.01

Notes: In **Counterfactual,** $C(n_{-1}, n)$ only the four adjustment cost parameters ($\gamma_f, \Gamma_f, \gamma_h$, and Γ_h) are kept at their 2006-2013 levels. All other parameters take their 2014-2019 values.

In **Counterfactual,** *free adjustments* all four adjustment costs parameters ($\gamma_f, \Gamma_f, \gamma_h$, and Γ_h) are set to zero. All other parameters take their 2014-2019 values.

In **Counterfactual,** α only the curvature of the revenue function (α) is kept at its 2006-2013 levels. All other parameters take their 2014-2019 values.

D.3 Proof of variance decomposition

From the reduced form revenue function derived initially, log labour productivity can be expressed as $\log(R) - \log(n) \equiv \log(A) + (\alpha - 1)\log(n) + \alpha \log(h)$. The variance of the sum is a sum of variances and covariances, such that:

$$\begin{aligned} &\text{Var}\big(\log(R) - \log(n)\big) = \text{Var}\big(\log(A)\big) + (\alpha-1)^2\,\text{Var}\big(\log(n)\big) + \alpha^2\,\text{Var}\big(\log(h)\big) \\ &+ 2\Big((\alpha-1)\,\text{Cov}\big(\log(A), \log(n)\big) + \alpha\,\text{Cov}\big(\log(A), \log(h)\big) + \alpha(\alpha-1)\,\text{Cov}\big(\log(n), \log(h)\big)\Big) \end{aligned} \tag{D1}$$

According to equation (11) in the main text this expression above should correspond to the trace of a particular matrix, *i.e.* the product between a matrix parameterised by the curvature of the revenue function and the variance-covariance matrix between profitability, employment, and average hours. As the focus lies on the trace of that resulting matrix, I can abstract from all elements outside that main diagonal. For convenience of notation, say that logged variables are expressed with tildes, such that $\tilde{x} \equiv \log(x)$:

$$\begin{bmatrix} \text{Var}(\tilde{A}) + 2(\alpha-1)\,\text{Cov}(\tilde{A},\tilde{n}) + 2\alpha\,\text{Cov}(\tilde{A},\tilde{h}) & \cdot & \cdot \\ \cdot & (\alpha-1)^2\,\text{Var}(\tilde{n}) + 2\alpha(\alpha-1)\,\text{Cov}(\tilde{n},\tilde{h}) & \cdot \\ \cdot & \cdot & \alpha^2\,\text{Var}(\tilde{h}) \end{bmatrix} \tag{D2}$$

It is immediate to realise that the expression shown in equation (D1) matches the sum of all three elements of the matrix in (D2).

References

Ackerberg, D. A., Caves, K., and Frazer, G. (2015). Identification properties of recent production function estimators. *Econometrica*, 83(6):2411–2451.

Addison, J. T., Portugal, P., and Vilares, H. (2017). Unions and collective bargaining in the wake of the great recession: evidence from portugal. *British Journal of Industrial Relations*, 55(3):551–576.

Banco de Portugal Microdata Research Laboratory (2025). Central Balance Sheet Harmonized Panel.

Bond, S., Hashemi, A., Kaplan, G., and Zoch, P. (2021). Some unpleasant markup arithmetic: Production function elasticities and their estimation from production data. *Journal of Monetary Economics*, 121:1–14.

Card, D. and Cardoso, A. R. (2022). Wage flexibility under sectoral bargaining. *Journal of the European Economic Association*, 20(5):2013–2061.

Cooper, R., Gong, G., Hu, G., and Yan, P. (2026). Exporting like china: Productivity, market demand, and labor frictions. *International Economic Review*, 67(2):729–758.

De Loecker, J. and Eeckhout, J. (2018). Global market power. Technical Report 24768, National Bureau of Economic Research.

De Loecker, J. and Syverson, C. (2021). An industrial organization perspective on productivity. In *Handbook of industrial organization*, volume 4, pages 141–223.

De Ridder, M., Grassi, B., and Morzenti, G. (2026). The hitchhiker's guide to markup estimation: assessing estimates from financial data. *Econometrica*, 94(1):137–168.

Decker, R. A., Haltiwanger, J., Jarmin, R. S., and Miranda, J. (2020). Changing business dynamism and productivity: Shocks versus responsiveness. *American Economic Review*, 110(12):3952–3990.

Santos, C. D., Costa, L. F., and Brito, P. B. (2022). Demand, supply and markup fluctuations. *The Economic Journal*, 132(644):1620–1645.

Statistics Portugal (2025). Inquérito Anual à Produção Industrial. dataset.

Varejão, J. and Portugal, P. (2007). Employment dynamics and the structure of labor adjustment costs. *Journal of Labor Economics*, 25(1):137–165.

Yeh, C., Macaluso, C., and Hershbein, B. (2022). Monopsony in the us labor market. *American Economic Review*, 112(7):2099–2138.